Locally Robust Policy Learning: Inequality, Inequality of Opportunity and Intergenerational Mobility*

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Abstract

Policy makers need to decide whether to treat or not to treat heterogeneous individuals. The optimal treatment choice depends on the welfare function that the policy maker has in mind and it is referred to as the policy learning problem. I study a general setting for policy learning with semiparametric Social Welfare Functions (SWFs) that can be estimated by locally robust/orthogonal moments based on U-statistics. This rich class of SWFs substantially expands the setting in Athey and Wager (2021) to non-additive SWFs that accommodate a wider range of distributional preferences. Three main applications of the general theory motivate the paper: (i) Inequality aware SWFs, (ii) Inequality of Opportunity aware SWFs and (iii) Intergenerational Mobility SWFs. I use the Panel Study of Income Dynamics (PSID) to assess the effect of attending preschool on adult earnings and estimate optimal policy rules based on parental years of education and parental income.

JEL Classification: C13; C14; C21; D31; D63; I24

Keywords: local robustness, U-statistics, Inequality, Intergenerational mobility, empirical welfare maximization.

R package (forthcoming): https://joelters.github.io/home/code/

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1 Introduction

Whenever a policy or treatment has heterogeneous effects it is important to decide carefully who should be treated. In the simplest case in which we care only about the average outcome, there are no budgetary limits and the treatment effect is positive for everyone, it follows that the best policy is to treat everyone. However, in most cases, we do not have such a luxury. We might have a limited budget, distributional concerns or negative treatment effects for some individuals. In these cases, it is important to decide whether *to treat or not to treat* different individuals. This is the problem of policy learning.

In economics we might want to know whether to provide training to the unemployed or design rules to assign conditional cash transfers. In business, we might want to know whether to provide a discount to a customer or whether to send price recommendations to some stores. Judges have to decide whether to release someone on parole. In education, we might want to know whether to provide a scholarship to a student or whether we should provide additional extra-curricular lessons. Certain medicines might be beneficial for some but detrimental for others or there might not be enough vaccines for the whole population.

The inherent distributional considerations in these examples are quite different. Hence, we need a framework accommodating different SWFs. The framework has to be general, but also needs to allow for estimation of optimal policies with certain statistical guarantees. I provide a framework to compute such optimal rules for a rich class of semiparametric SWFs, estimable by U-statistics. Leading examples are the average outcome, but also Inequality aware SWFs, Inequality of Opportunity (IOp) aware SWFs and Intergenerational Mobility SWFs.

To my knowledge, there is no prior work on IOp and Intergenerational mobility SWFs in the policy learning literature. IOp is the part of inequality explained by circumstances X outside the control of the individual, e.g. sex, race or parental income. IOp SWFs do not penalize all inequality, only unfair inequality (i.e. inequality explained by circumstances). Based on the seminal contributions in Van De Gaer (1993), Fleurbaey (1995) and Roemer (1998), the IOp literature has focused on çmeasuring IOp. A popular measure is the Gini of the best predictions (in mean squared error sense) of the outcome Y given the circumstances X, i.e. $G(\gamma(X))$ where $\gamma(X) = \mathbb{E}[Y|X]$ and $G(Z)$ denotes the Gini index of the random variable Z. To accommodate a possibly high-dimensional set of circumstances, IOp literature has started using machine learners to predict (e.g Brunori et al. (2019a), Brunori et al. (2019b), Brunori et al. (2021), Brunori and Neidhöfer (2021), Rodríguez et al. (2021), Carranza (2022) or Hufe et al. (2022)). The bias-variance trade-off in the prediction might allow for some bias which creeps into the IOp estimator. Escanciano and Terschuur (2023) provide locally robust IOp estimators robust to such biases. I use these results to construct Neyman-orthogonal IOp aware SWFs.

Inequality SWFs have been studied in Kasy (2016), Kitagawa and Tetenov (2021) or Kock et al. (2023). A popular SWF for an outcome Y is $W = \mathbb{E}[Y](1-G(Y))$. This SWF values the average outcome but penalizes high inequality (the Gini is between 0 and 1). This is a particular case of my setting, as the Gini coefficient is a ratio of U-statistics.

Leqi and Kennedy (2021) propose to maximize average conditional quantiles. Cui and Han (2024) focus on a conditional quantile of the treatment effect using partial identification. Wang et al. (2018) study quantile-optimal policies and adapt their theory to minimize Gini's mean difference. They use the U-statistics nature of the Gini mean difference and obtain asymptotic theory for a particular class of policy rules by using empirical U-process methods. I avoid U-processes theory by using a representation of U-statistics as sums-of-i.i.d. blocks introduced in Hoeffding (1963). This representation is key in proving the main result of the paper.

While inequality aware SWFs look at the distribution of Y, IOp aware SWFs focus on the distribution of the predictions $\gamma(X)$. A natural IOp aware SWF is $W = \mathbb{E}[Y](1 - G(\gamma(X)))$, which only penalizes IOp. This example adds an extra nuisance parameter, $\gamma(X)$, on top of the conditional expectations/propensity scores needed to identify treatment effects. Policy learning with general semiparametric SWFs, which directly depend on additional unknown functions beyond those required for identifying treatment effects, has been little explored, with the exception of Leqi and Kennedy (2021) whose welfare depends on conditional quantiles.

Intergenerational mobility studies the relationship between child and parental outcomes. The Kendall-τ is a popular measure of mobility in the literature (see Chetty et al. (2014) or Kitagawa et al. (2018)). It looks at whether the parents of individual i are richer than those of j and i is richer than j. An intergenerational mobility aware SWF is $W = -|\tau - t|$ for some target $t \in [-1, 1]$. For instance, we could decide the allocation of higher education scholarships to students based on their characteristics whenever the effect of education on long-term income can be identified and we want to reduce dependence between parental and child's income.

The policy learning literature, sometimes called empirical welfare maximization or offline policy learning, looks for optimal allocation rules π mapping individual characteristics to binary treatment decisions. Optimal rules are searched within a class Π of treatment rules to maximize welfare. Following Manski (2004), I search for optimal policies in Π so as to minimize regret, i.e. the expected difference between the best possible welfare and the welfare evaluated at the estimated policy. Other relevant work includes Dehejia (2005), Hirano and Porter (2009), Stoye (2009, 2012), Chamberlain (2011), Bhattacharya and Dupas (2012), Tetenov (2012), Kasy (2016), Kitagawa and Tetenov (2018, 2021), Athey and Wager (2021), or Zhou et al. (2023).

Non-parametric estimation of unknown functions in semiparametric SWFs poses a challenge to the statistical guarantees of estimated policy rules. This stems from the slow convergence of non-parametric estimators, addressed in semiparametric methods through locally robust/orthogonal moments. These are alternative moment conditions that identify the quantity of interest and allow for its estimation at $\sqrt{n}$ (where n is the sample size) rate. I expand previous work by considering any semiparametric SWF, possibly defined as a U-statistic, which can be estimated by locally robust/orthogonal scores. The main theoretical result provides an asymptotic upper bound to the regret of the estimated policy rule.

This paper is related to Athey and Wager (2021), Leqi and Kennedy (2021) and Zhou et al. (2023) in making use of the semiparametric literature on locally robust/orthogonal scores (e.g. Chernozhukov et al. (2022)) to obtain $\sqrt{n}$ rates of convergence even with nonparametric first steps. I build upon Escanciano and Terschuur (2023) to expand policy learning results to SWFs defined by U-statistics. Athey and Wager (2021) find rates of the regret that optimally depend on the complexity Π and the sample size in observational settings where the propensity score is unknown. They do so for average-treatment-like SWFs. I generalize this setting by allowing arbitrary semiparametric SWFs, possibly defined as U-statistics, as long as they can be identified by locally robust/orthogonal scores.

Empirically, treatment allocation with inequality, IOp and IGM SWFs is hard. We need circumstances and parental income which are usually absent and to identify treatment effects. I look at the effect of preschool on adult earnings using the Panel Study of Income Dynamics (PSID) dataset. This empirical illustration has many advantages. Any variable that induces preschool attendance can be considered a circumstance under the (very reasonable) assumption that we cannot hold the kid responsible for these variables. Also, PSID has been following families for nearly 50 years meaning we have rich information on family background. Further, PSID allows us to look at long-term outcomes such as adult earnings.

The empirical illustration has limitations. Treatment is not randomly assigned, so identifying treatment effects relies on the assumption of selection on observables. Preschool attendance is not a binary treatment since its quality varies. Furthermore, I have no information on the cost of treatment and allocating children to preschool based on their circumstances might not be enforceable or ethical. Observed preschool choices differ from estimated optimal rules, even when maximizing average outcomes, suggesting parents prioritize factors beyond future earnings.

The effect of preschool is heterogeneous. On average preschool has a positive effect on adult earnings but children with highly educated mothers and high parental income are negatively affected. This aligns with findings in psychology and economics documenting that in early education, there is less interaction with adults and hence, a negative effect of attending such institutions if the interactions with adults at home are of "higher-quality" (see Fort et al. (2020)).

These heterogeneous effects have different implications when estimating optimal treatment rules for different SWFs. I compute optimal treatment rules based on parental income and mother's education in a class of decision trees of depth two. Estimated optimal rules maximizing the average try to treat anyone with a positive treatment effect. Inequality aware SWFs also treat individuals with negative treatment effects since the decrease in inequality compensated the decrease in average earnings. The same happens with the IGM welfare which has no average motive at all and only cares about decreasing the association between parental and child's income. The additive and IOp estimated optimal policy rules coincide while the inequality and IGM estimated rules treat individuals who do not benefit from treatment or do not treat individuals who do benefit from treatment. The coincidence of the additive and IOp rules is specific to the heterogeneous treatment effects in the data and is not a general result. In this empirical illustration, we can see that this happens because maximizing the average already decreases IOp drastically.

I introduce the welfare objects which serve as guiding examples in Section 2. Section 3 elaborates on the general theory for SWFs identified by locally robust/orthogonal scores which are linear on the distribution of the data (i.e. not defined as U-statistics) and Section 4 generalized to SWFs possibly defined as U-statistics. Section 5 provides upper bounds on the regret and Section 6 deals with the empirical illustration. All proofs are in the Appendix.

2 Welfare economics for inequality, IOp and rank correlations

The policy learning literature is at the intersection of welfare economics and econometrics. Before we delve into the econometric problem I present the main welfare objects we are going to be interested in. Suppose we have some continuous random outcome $Y \in \mathbb{R}^+$. The most basic SWF is the additive welfare based on the average outcome

$$W = \mathbb{E}[Y].$$

The above welfare does not care about distributional aspects apart from the average. A first approach to include distributional concerns in our analysis is to follow Dalton (1920) and Atkinson et al. (1970) and consider increasing and concave transformations $u(\cdot)$ [1]

$$W = \mathbb{E}[u(Y)].$$

This SWF will rank two outcome distributions equally for all increasing and concave $u(\cdot)$ if the Lorenz curve of one of the distributions is everywhere above the Lorenz curve of the other distribution and has equal or higher mean; equivalently if one distribution second-order stochastically dominates the other. If we want to obtain a complete ordering we need to specify $u(\cdot)$ further. One popular choice is

$$u(y) = \begin{cases} \frac{y^{1-\theta}}{1-\theta} & \text{if } \theta \in (0,1) \\ \log(y) & \text{if } \theta = 1, \end{cases}$$

where θ captures the concavity of $u(\cdot)$ and can be interpreted as an inequality aversion parameter. I also focus on SWFs aware of Inequality of Opportunity (IOp). IOp is the part of total inequality which can be explained by circumstances, i.e. by variables that are outside the control of the individual such as parental education or parental income. Let $X \in \mathbb{R}^k$ be such a random vector of circumstances. Let also $\gamma(X) = \mathbb{E}[Y|X]$. By looking at the distribution of $\gamma(X)$ instead of that of the outcome Y we get IOp averse SWFs. For instance,

$$W = \mathbb{E}[u(\gamma(X))].$$

[1] With abuse of notation I call W to all SWFs as they appear.

If there is no IOp, circumstances are unable to predict the outcome and we have $\gamma(X) = \mathbb{E}[Y]$. In this case, $W = u(\mathbb{E}[Y])$ so we only care about the (transformed) average income. If we have maximum IOp, the outcome is a deterministic function of the circumstances and $\gamma(X) = Y$. Then, $W = \mathbb{E}[u(Y)]$. Since all inequality is IOp, we are back to the inequality averse SWF. Alternatively, we can weigh differently different parts of the distribution. Let F_Y be the distribution of Y and F_Y^{-1} be the quantile function. For weights $w(\cdot)$, we might have

$$W = \int_0^1 F_Y^{-1}(\tau)w(\tau)d\tau.$$

This welfare has been used in Mehran (1976), Donaldson and Weymark (1980), Weymark (1981), Donaldson and Weymark (1983) or Aaberge et al. (2021). Letting $w_k(\tau) = (k-1)(1-\tau)^{k-2}$ we get the extended Gini family of SWFs. I focus on $k = 3$, the standard Gini SWF which can be shown to be

$$\begin{aligned} W &= \mathbb{E}[Y](1 - G(Y)) \\ &= (1/2)\mathbb{E}[Y_i + Y_j - |Y_i - Y_j|], \end{aligned}$$

where Y_i and Y_j are independent and distributed as F_Y. The second equality follows from writing the Gini of Y as $G(Y) = \mathbb{E}[|Y_i - Y_j|]/\mathbb{E}[Y_i + Y_j]$ (i.e. the Gini can be interpreted as a normalized expected absolute distance between the outcomes of two individuals taken at random). The welfare above is additive if there is no inequality and penalizes positive values of the Gini coefficient. If we do not care about inequality but only about IOp we can look at the distribution of $\gamma(X)$ instead of the distribution of Y. In that case, we have

$$\begin{aligned} W &= \mathbb{E}[\gamma(X)](1 - G(\gamma(X))) \\ &= (1/2)\mathbb{E}[\gamma(X_i) + \gamma(X_j) - |\gamma(X_i) - \gamma(X_j)|], \end{aligned}$$

where X_i and X_j are independent and distributed as X. If $G(\gamma(X)) = 0$, we are back to the additive case. If there is full IOp, then $G(\gamma(X)) = G(Y)$ and we are back to the standard Gini SWF. Finally, I also consider the problem of intergenerational mobility. Let the first component of X, $X_1 \in \mathbb{R}$, be the parental outcome. A measure of association between Y and X_1 is the Kendall-τ

$$\tau = \mathbb{E}[sgn(Y_i - Y_j)sgn(X_{1i} - X_{1j})],$$

where $sgn(a) = \mathbb{1}(a > 0) - \mathbb{1}(a < 0)$. This parameter is popular in the intergenerational mobility literature (see Chetty et al. (2014) or Kitagawa et al. (2018)). It takes values between 1 and -1. $\tau = 1$ means that whenever an individual has a higher income than another, she also has a higher parental income. $\tau = -1$ is the opposite. For some target Kendall-τ $t \in [-1, 1]$ an intergenerational mobility aware SWF is

$$W = -\left|\mathbb{E}\left[sgn(Y_i - Y_j)sgn(X_{1i} - X_{1j})\right] - t\right|.$$

To my knowledge, this is a novel SWF. Note that it allows us to treat problems much more general than intergenerational mobility. Setting $t = 0$, maximizing this SWF corresponds to allocating a treatment to minimize the dependence between two variables Y and X_1.

3 Policy learning with general orthogonal scores

Let $(Y(1), Y(0), D, X) \sim F_0$ where $(Y(1), Y(0)) \in \mathcal{Y} \times \mathcal{Y} \subseteq \mathbb{R} \times \mathbb{R}$ are potential outcomes, i.e. $Y(1)$ is the outcome under treatment and $Y(0)$ is the outcome in the absence of treatment. D is a binary treatment and $X \in \mathcal{X}$ is a vector of pre-treatment covariates. Let $\gamma^{(j)}(X) = \mathbb{E}[Y(j)|X] \in \mathbb{R}$ for $j = 0, 1$ be potential predictions, i.e. the predictions of the potential outcomes given X. We observe only realizations of (Y, D, X) where $Y = Y(1)D + Y(0)(1-D)$. Let $\pi : \mathcal{X} \mapsto \{0, 1\}$ be a binary treatment rule and Π be a collection of such treatment rules. Let $g : \mathcal{Y} \times \mathcal{X} \times \mathbb{R} \to \mathbb{R}$ be a measurable function. We are interested in choosing $\pi \in \Pi$ to maximize

$$W(\pi) = \mathbb{E}[g(Y(1), X, \gamma^{(1)})\pi(X) + g(Y(0), X, \gamma^{(0)})(1 - \pi(X))], \tag{3.1}$$

where $g(Y(j), X, \gamma^{(j)})$ is shorthand for $g(Y(j), X, \gamma^{(j)}(X))$ for $j = 0, 1$. For additive welfare, $g(Y(j), X, \gamma^{(j)}) = Y(j)$ for $j = 0, 1$. Importantly, g can depend on possibly infinite-dimensional nuisance parameters $\gamma^{(j)}$. While throughout the paper I consider $\gamma^{(j)}$ to be a conditional expectation, this framework can be extended to allow for much more general first steps such as conditional quantiles (see Ichimura and Newey (2022)).

Example 1 (IOp Atkinson) *For an inequality averse SWF we can set* $g(y, x, \gamma^{(j)}) = u(y)$ *with* $u(\cdot)$ *a concave function. Then,* $W(\pi) = \mathbb{E}[u(Y(1))\pi(X) + u(Y(0))(1 - \pi(X))]$*. In this case, the methods in Kitagawa and Tetenov (2018) and Athey and Wager (2021) can be applied. Instead, for an IOp averse SWF we look at the distribution of* $\gamma^{(j)}(X)$ *so we set* $g(y, x, \gamma^{(j)}) = u(\gamma^{(j)}(x))$ *for* $j = 0, 1$*. Then,*

$$W(\pi) = \mathbb{E}[u(\gamma^{(1)}(X))\pi(X) + u(\gamma^{(0)}(X))(1 - \pi(X))].$$

This welfare has not been covered in the policy learning literature before. ■

(3.1) is not observable since for a given individual we do not observe both potential outcomes. To identify (3.1) we need our sample to come from an experimental or observational experiment where the policy has already been implemented. Let $e(X) = \mathbb{P}(D = 1|X)$ be the propensity score. I assume that the following holds.

Assumption 1 *i)* $(Y(1), Y(0)) \perp D|X$*, ii) There exists* $\kappa \in (0, 1/2]$ *such that* $e(x) \in [\kappa, 1 - \kappa]$*.*

Now I state the first identification result. With some abuse of notation let $\gamma(d, x) = \mathbb{E}[Y|D = d, X = x]$ and notice that under Assumption 1, $\gamma(j, X) = \gamma^{(j)}(X)$ a.s. for $j = 0, 1$. Define also $\varphi(d, x, \gamma) = \mathbb{E}[g(Y, x, \gamma(d, x))|D = d, X = x]$.

Proposition 3.1 *Under Assumption 1,* $W(\pi)$ *is identified as*

$$W(\pi) = \mathbb{E}[\varphi(1, X, \gamma)\pi(X) + \varphi(0, X, \gamma)(1 - \pi(X))].$$

Note that if g does not depend on potential outcomes directly, i.e. $g(u, X, \gamma^{(j)}) = g(t, X, \gamma^{(j)}) \equiv g(X, \gamma^{(j)})$ for all $u, t \in \mathcal{Y}$ then φ is known since $\varphi(D, X, \gamma) = g(X, \gamma(D, X))$. This happens in all IOp examples such as Example 1 and Example 3 below. Hence, we can have either γ or (γ, φ) as nuisance parameters. Define linear sets $\mathcal{G}$ and $\mathcal{F}$ where γ and φ live respectively. For any $\tilde{\gamma} \in \mathcal{G}$, let $\gamma_\tau = \gamma + \tau\tilde{\gamma}$. To enjoy local robustness to high dimensional and ML first steps, I now provide orthogonal scores. I need the following assumption and lemma

Assumption 2 *There exist functions* $\alpha_1 \in \mathcal{A}$ *and* $\alpha_0 \in \mathcal{A}$ *such that for any* $\tilde{\gamma} \in \mathcal{G}$

with $\mathbb{E}[\tilde{\gamma}(D,X)^2] < \infty$

$$\frac{d}{d\tau}\mathbb{E}\left[\frac{D}{e(X)}g(Y,X,\gamma_\tau)\pi(X)\right]\bigg|_{\tau=0} = \frac{d}{d\tau}\mathbb{E}[\alpha_1(D,X)\gamma_\tau(D,X)\pi(X)]\bigg|_{\tau=0},$$
$$\frac{d}{d\tau}\mathbb{E}\left[\frac{1-D}{1-e(X)}g(Y,X,\gamma_\tau)\pi(X)\right]\bigg|_{\tau=0} = \frac{d}{d\tau}\mathbb{E}[\alpha_0(D,X)\gamma_\tau(D,X)\pi(X)]\bigg|_{\tau=0}.$$

and $\mathbb{E}[\alpha_j(D,X)^2] < \infty$ *for* $j = 0, 1$.

This is a common assumption in the semiparametric literature (e.g. (4.1) in Newey (1994)) and allows for non-linear dependence on γ, generalizing Assumption 1 in Athey and Wager (2021). The next Lemma is useful for deriving the orthogonal scores.

Lemma 1 *Under Assumption 2, for* $j = 0, 1$

$$\frac{d}{d\tau}\mathbb{E}[\varphi(j,X,\gamma_\tau)\pi(X)]\bigg|_{\tau=0} = \frac{d}{d\tau}\mathbb{E}[\alpha_j(D,X)\gamma_\tau(D,X)\pi(X)]\bigg|_{\tau=0}.$$

Define now propensity score weights $r_1(d,x) = d/e(x)$ and $r_0(d,x) = (1-d)/(1-e(x))$, with r_1 and r_2 living in a linear set $\mathcal{R}$. The orthogonal scores are given by the next Proposition.

Proposition 3.2 *The orthogonal score is given by*

$$\Gamma(\pi) = \Gamma_1\pi(X) + \Gamma_0(1-\pi(X)), \tag{3.2}$$

where

$$\Gamma_1 = \varphi(1,X,\gamma) + r_1(D,X)(g(Y,X,\gamma(D,X)) - \varphi(D,X,\gamma)) + \alpha_1(D,X)(Y-\gamma(D,X)),$$
$$\Gamma_0 = \varphi(0,X,\gamma) + r_0(D,X)(g(Y,X,\gamma(D,X)) - \varphi(D,X,\gamma)) + \alpha_0(D,X)(Y-\gamma(D,X)).$$

Orthogonal scores are formed by identifying scores and correction terms for nuisance parameters φ and γ. Whenever g does not depend on the potential outcomes directly we have that $g(Y,X,\gamma(D,X)) - \varphi(D,X,\gamma) = 0$ a.s.

3.1 Double robustness

As known for the Augmented Inverse Probability Weighting (AIPW) (Robins et al. (1994)), we could have started by identifying $W(\pi)$ using an inverse propensity score weighting (IPW) approach to get

$$W(\pi) = \mathbb{E}\Big[\frac{Dg(Y,X,\gamma(D,X))}{e(X)}\pi(X) + \frac{(1-D)g(Y,X,\gamma(D,X))}{1-e(X)}(1-\pi(X))\Big].$$

Under Assumption 2 and treating e as a nuisance parameter instead of γ, we end up with the same orthogonal score (3.2). In fact, as with the AIPW, the orthogonal scores are a combination of IPW and regression based scores which yield attractive double robustness properties. Define for $j = 0, 1$

$$\begin{aligned}\psi_j(\varphi, r_j, \gamma, \alpha_j) = \mathbb{E}[&\varphi(j, X, \gamma) + r_j(D,X)(g(Y,X,\gamma(D,X)) - \varphi(D,X,\gamma)) \\ &+ \alpha_j(D,X)(Y - \gamma(D,X))].\end{aligned}$$

Proposition 3.3 *For any measurable functions $\bar{\varphi} \in \mathcal{F}, \bar{r}_j \in \mathcal{R}$, $\bar{\alpha}_j \in \mathcal{A}_j$ for $j = 0, 1$ we have that*

$$\psi_j(\bar{\varphi}, \bar{r}_j, \gamma, \bar{\alpha}_j) = \psi_j(\varphi, r_j, \gamma, \alpha_j) + \mathbb{E}\left[(r_j(D,X) - \bar{r}_j(D,X))(\bar{\varphi}(D,X,\gamma) - \varphi(D,X,\gamma))\right].$$

Furthermore, if $g(y, x, \gamma(d, x))$ is affine in its third argument we also have that for any $\bar{\gamma} \in \mathcal{G}$

$$\begin{aligned}\psi_j(\bar{\varphi}, \bar{r}_j, \bar{\gamma}, \bar{\alpha}_j) = \psi_j(\varphi, r_j, \gamma, \alpha_j) &+ \mathbb{E}\left[(r_j(D,X) - \bar{r}_j(D,X))(\bar{\varphi}(D,X,\bar{\gamma}) - \varphi(D,X,\bar{\gamma}))\right] \\ &+ \mathbb{E}\left[(\alpha_j(D,X) - \bar{\alpha}_j(D,X))(\bar{\gamma}(D,X) - \gamma(D,X))\right].\end{aligned}$$

In other words, as long as γ is correctly specified, we are robust to misspecification of α_j and it is enough to have either φ or r_j correctly specified. Furthermore, if g is affine in its third argument we have multiple robustness in the sense that it is enough to have either φ or r_j correctly specified and either γ or α_j correctly specified.

3.2 Estimation

Suppose we have an i.i.d. sample (Y_i, D_i, X_i) for $i = 1, ..., n$ distributed as (Y, D, X). To estimate the welfare for a given $\pi \in \Pi$ we employ cross-fitting as in Chernozhukov et al. (2022). Let the data be split in L groups $I_1, ..., I_l$, then

$$\hat{W}_n(\pi) = \frac{1}{n}\sum_{l=1}^{L}\sum_{i \in I_l} \hat{\Gamma}_{1i,l}\pi(X_i) + \hat{\Gamma}_{0i,l}(1 - \pi(X_i)),$$

where

$$\hat{\Gamma}_{1i,l} = \hat{\varphi}_l(1, X_i, \hat{\gamma}_l) + \frac{D_i}{\hat{e}_l(X_i)}(Y_i - \hat{\varphi}_l(D_i, X_i, \hat{\gamma}_l)) + \hat{\alpha}_{1,l}(D_i, X_i)(Y_i - \hat{\gamma}_l(D_i, X_i)),$$
$$\hat{\Gamma}_{0i,l} = \hat{\varphi}_l(0, X_i, \hat{\gamma}_l) + \frac{1 - D_i}{1 - \hat{e}_l(X_i)}(Y_i - \hat{\varphi}_l(D_i, X_i, \hat{\gamma}_l)) + \hat{\alpha}_{0,l}(D_i, X_i)(Y_i - \hat{\gamma}_l(D_i, X_i)),$$

and $(\hat{\varphi}_l, \hat{e}_l, \hat{\gamma}_l, \hat{\alpha}_{j,l})$, $j = 0, 1$, are estimators of the nuisance functions which do not use observations in I_l.

Example 1 (IOp Atkinson (cont.)) *For $\theta \in (0, 1]$, let*

$$U(\gamma(x)) = \begin{cases} \frac{\gamma(x)^{1-\theta}}{1-\theta} & \text{if } \theta \in (0,1) \\ \log(\gamma(x)) & \text{if } \theta = 1. \end{cases}$$

In this case, $g = U$ depends only on nuisance parameters. The orthogonal score for $\theta \in (0, 1]$ is

$$\Gamma_i(\pi) = U(\gamma(1, X_i)) + \frac{\gamma(D_i, X_i)^{-\theta} D_i}{e(X_i)}(Y_i - \gamma(D_i, X_i))\pi(X_i)$$
$$+ U(\gamma(0, X_i)) + \frac{\gamma(D_i, X_i)^{-\theta}(1 - D_i)}{1 - e(X_i)}(Y_i - \gamma(D_i, X_i))(1 - \pi(X_i)),$$

i.e. $\alpha_1(D_i, X_i) = e(X_i)^{-1}\gamma(D_i, X_i)^{-\theta} D_i$ and $\alpha_0(D_i, X_i) = (1-e(X_i))^{-1}\gamma(D_i, X_i)^{-\theta}(1-D_i)$. ■

The estimator of the optimal treatment rule among a class of rules Π is

$$\hat{\pi} = \arg\max_{\pi \in \Pi} \hat{W}_n(\pi).$$

4 Policy learning with U-statistics

Let now $\pi_{ab}(X_i, X_j) = \mathbb{1}(\pi(X_i) = a) \times \mathbb{1}(\pi(X_j) = b)$ with $a, b \in \{0, 1\}$. Our most general welfare takes the form

$$W(\pi) = \mathbb{E}\Big[\sum_{(a,b)\in\{0,1\}^2} g(Y_i(a), X_i, Y_j(b), X_j, \gamma^{(a)}, \gamma^{(b)})\pi_{ab}(X_i, X_j)\Big]. \qquad (4.1)$$

Abusing notation, we keep using g in this pairwise setting. g takes now more arguments compared to previous sections. This is because $W(\pi)$ depends on expected pairwise comparisons. For instance, in the Gini, we look at the expected absolute distance between two individuals taken at random. Also, we are summing across $\{0,1\}^2$. This is because we have to take into account when both members of the pair are under treatment, or just one of them or none of them. Finally, we have $\pi_{ab}(X_i, X_j)$ instead of $\pi(X_i)$ to indicate when both members of the pair are allocated to treatment, just one of them or none of them. Also, whenever g does not depend on some of its arguments I omit them.

Example 2 (Inequality) *We can accommodate the standard Gini SWF with*

$$g(Y_i(a), Y_j(b)) = (1/2)(Y_i(a) + Y_j(b) - |Y_i(a) - Y_j(b)|).$$

■

Example 3 (Inequality of Opportunity IOp) *We can apply the standard Gini SWF to the distribution of the predictions to get* $\mathbb{E}[\gamma(X_i)](1 - G(\gamma(X_i)))$. *This fits our setting by letting*

$$g(X_i, X_j, \gamma^{(a)}, \gamma^{(b)}) = (1/2)(\gamma^{(a)}(X_i) + \gamma^{(b)}(X_j) - |\gamma^{(a)}(X_i) - \gamma^{(b)}(X_j)|).$$

■

Example 4 (Kendal-τ) *To allocate a treatment targeting a specific Kendall-τ, say $t \in \mathbb{R}$, we have to extend our setting to transformations of the right-hand side of 4.1*

$$g(Y_i(a), X_{1i}, Y_j(b), X_{1j}) = sgn(Y_i(a) - Y_j(b))sgn(X_{1i} - X_{1j}),$$
$$W(\pi) = -\left|\mathbb{E}\left[\sum_{(a,b)\in\{0,1\}^2} g(Y_i(a), X_{1i}, Y_j(b), X_{1j})\pi_{ab}(X_i, X_j)\right] - t\right|.$$

■

For $a, b \in \{0,1\}$ let now $\varphi(a, X_i, b, X_j, \gamma) = \mathbb{E}[g(Y_i, X_i, Y_j, X_j, \gamma(a, X_i), \gamma(b, X_j))|D_i = a, X_i, D_j = b, X_j]$.

Proposition 4.1 *Under Assumption 1, $W(\pi)$ in (4.1) is identified in the following way*

$$W(\pi) = \mathbb{E}\Big[\sum_{(a,b)\in\{0,1\}^2} \varphi(a, X_i, b, X_j, \gamma)\pi_{ab}(X_i, X_j).\Big],$$

Now I apply Proposition 4.1 to identify the welfare in each of our three main examples.

Example 2 (Inequality (cont.)) *In this example, welfare is identified by*

$$W(\pi) = \mathbb{E}\Big[\frac{1}{2}\sum_{(a,b)\in\{0,1\}^2} \mathbb{E}(Y_i + Y_j - |Y_i - Y_j| \mid D_i = a, X_i, D_j = b, X_j)\pi_{ab}(X_i, X_j)\Big].$$

■

Example 3 (IOp (cont.)) *In this example, welfare is identified by*

$$W(\pi) = \frac{1}{2}\mathbb{E}\Big[\sum_{(a,b)\in\{0,1\}^2} \Big(\gamma(a, X_i) + \gamma(b, X_j) - |\gamma(a, X_i) - \gamma(b, X_j)|\Big)\pi_{ab}(X_i, X_j)\Big].$$

■

Example 4 (Intergenerational mobility (cont.)) *In this example, welfare is identified by*

$$W(\pi) = -\Big|\mathbb{E}\Big[\frac{1}{2}\sum_{(a,b)\in\{0,1\}^2} \mathbb{E}(sgn(X_{1i}-X_{1j})sgn(Y_i-Y_j) \mid D_i = a, X_i, D_j = b, X_j)\pi_{ab}(X_i, X_j)\Big]-t\Big|.$$

■

Example 3 does not depend on the potential outcomes which makes the expression simpler. To compute orthogonal scores we need to assume a linearization property as in Assumption 2. We assume directly the result in Lemma 1 extended to the pairwise setting for brevity. However, I could also assume a sufficient condition analogous to Assumption 2

Assumption 3 *There exist $\alpha_{ab,p} \in \mathcal{A}$, $P < \infty$, and (c_{1p}, c_{2p}) for $p = 1, ..., P$, such that for all $(a, b) \in \{0, 1\}^2$ the following linearization holds*

$$\frac{d}{d\tau}\mathbb{E}[\varphi(a, X_i, b, X_j, \gamma_\tau)] = \frac{d}{d\tau}\mathbb{E}\Big[\sum_{p=1}^{P} \alpha^{\gamma}_{ab,p}(D_i, X_i, D_j, X_j)(c_{1p}\gamma_\tau(D_i, X_i)+c_{2p}\gamma_\tau(D_j, X_j))\Big],$$

where γ_τ is defined as in Assumption 2 and $\mathbb{E}[\alpha_j(D_i, X_i)^2] < \infty$.

The orthogonal scores are given by the next Proposition.

Proposition 4.2 *The orthogonal scores are given by*

$$\Gamma_{ij}(\pi) = \sum_{(a,b)\in\{0,1\}^2} \Gamma_{ij}^{ab}\pi_{ab}(X_i, X_j),$$

where

$$\Gamma_{ij}^{ab} = \varphi(a, X_i, b, X_j, \gamma) + \phi_{ab}^{\varphi}(D_i, X_i, D_j, X_j, \varphi, \alpha^{\varphi}) + \phi_{ab}^{\gamma}(D_i, X_i, D_j, X_j, \gamma, \alpha^{\gamma}),$$

$$\phi_{ab}^{\gamma}(D_i, X_i, D_j, X_j, e, \alpha^{\gamma}) = \sum_{p=1}^{P} \alpha_{ab,p}^{\gamma}(D_i, X_i, D_j, X_j)(c_{1p}Y_i + c_{2p}Y_j - c_{1p}\gamma(D_i, X_i) - c_{2p}\gamma(D_j, X_j)),$$

$$\phi_{ab}^{\varphi}(D_i, X_i, D_j, X_j, \varphi, \alpha^{m}) = \alpha_{ab}^{\varphi}(D_i, X_i, D_j, X_j)(g(Y_i, X_i, Y_j, X_j, \gamma(a, X_i), \gamma(b, X_j)) - \varphi(D_i, X_i, D_j, X_j, \gamma)),$$

and

$$\alpha_{ab}^{\varphi}(D_i, X_i, D_j, X_j) = \frac{D_{ij}^{ab}}{e_{ab}(X_i, X_j)}, \quad D_{ij}^{ab} = \mathbb{1}(D_i = a)\mathbb{1}(D_j = b), \quad e_{ab}(X_i, X_j) = \mathbb{E}[D_{ij}^{ab}|X_i, X_j].$$

Again, whenever g does not directly depend on potential outcomes, $g = \varphi$ and $\phi_{ab}^{\varphi} = 0$. As before we have two nuisance parameters φ and γ with associated nuisance parameters denoted with α. This orthogonal score also can be arrived at starting from an IPW identification with γ and e_{ab} as nuisance parameters. As before, this score enjoys the double robustness properties as in Proposition 3.3 from combining IPW and regression based approaches. Even though these robustness properties are novel in the U-statistic setting, the computations are analogous to those in the proof of Proposition 3.3 and I omit them here for brevity.

Example 2 (Inequality (cont.)) *In this example, we have that*

$$\begin{aligned}\Gamma_{ij}^{ab} &= \frac{1}{2}\mathbb{E}(Y_i + Y_j - |Y_i - Y_j| \mid D_i = a, X_i, D_j = b, X_j) \\ &\quad + \frac{D_{ij}^{ab}}{2e_{ab}(X_i, X_j)}(Y_i + Y_j - |Y_i - Y_j| - \mathbb{E}(Y_i + Y_j - |Y_i - Y_j| \mid D_i = a, X_i, D_j = b, X_j)).\end{aligned}$$

■

Example 3 (IOp (cont.)) *I introduce the orthogonal score for IOp example as a Proposition.*

Proposition 4.3 *Assume for all* $(a,b) \in \{0,1\}^2$ *that either (i)* $\mathbb{P}(\gamma_(a,X_i)-\gamma(b,X_j)=0)=0$ *or that (ii)* $x_i \neq x_j \implies \gamma(a,X_i)-\gamma(b,X_j)\neq 0$ *and let* $\delta_{ij}^{ab} = sgn(\gamma(a,X_i)-\gamma(b,X_j))$*, then*

$$\Gamma_{ij}^{ab} = \frac{1}{2}\Bigg(\gamma(a,X_i)+\gamma(b,X_j)-|\gamma(a,X_i)-\gamma(b,X_j)| \\ +\frac{\mathbb{1}(D_i=a)}{e_a(X_i)}(1-\delta_{ij}^{ab})(Y_i-\gamma(D_i,X_i))+\frac{\mathbb{1}(D_j=b)}{e_b(X_j)}(1+\delta_{ij}^{ab})(Y_j-\gamma(D_j,X_j))\Bigg).$$

These assumptions deal with the point of non-differentiability of the absolute value. For a thorough discussion see Escanciano and Terschuur (2023). ■

To estimate the welfare for a given $\pi \in \Pi$, I use a cross-fitting algorithm used in Escanciano and Terschuur (2023). I split the pairs $\{(i,j) \in \{1,...,n\}^2 : i < j\}$ in L groups $I_1,...,I_l$, then

$$\hat{W}_n(\pi) = \binom{n}{2}^{-1}\sum_{l=1}^{L}\sum_{(i,j)\in I_l}\hat{\Gamma}_{ij,l}(\pi), \tag{4.2}$$

where $\hat{\Gamma}_{ij,l}$ is the same as Γ_{ij} but with all nuisance parameters replaced by estimators which do not use observations in I_l. The estimator of the optimal treatment rule is

$$\hat{\pi} = \arg\max_{\pi\in\Pi}\hat{W}_n(\pi).$$

For the Intergenerational mobility example, the estimation is slightly different.

Example 4 (Intergenerational mobility (cont.)) *The orthogonal score is given by*

$$\Gamma_{ij}^{ab} = \mathbb{E}(sgn(X_{1i}-X_{1j})sgn(Y_i-Y_j) \mid D_i=a, X_i, D_j=b, X_j) \\ +\frac{D_{ij}^{ab}}{e_{ab}(X_i,X_j)}(sgn(X_{1i}-X_{1j})sgn(Y_i-Y_j)-\mathbb{E}(sgn(X_{1i}-X_{1j})sgn(Y_i-Y_j) \mid D_i=a, X_i, D_j=b, X_j)).$$

The estimator of the welfare for a given $\pi \in \Pi$ *and target* t *is*

$$\hat{W}_n(\pi) = -\left|\binom{n}{2}^{-1}\sum_{l=1}^{L}\sum_{(i,j)\in I_l}\sum_{(a,b)\in\{0,1\}^2}\hat{\Gamma}_{ij,l}^{ab}\pi_{ab}(X_i,X_j)-t\right|. \tag{4.3}$$

■

5 Asymptotic statistical guarantees

Now it is useful to make clear the dependence of the scores Γ_{ij}^{ab} on the data and the nuisance parameters. Hence, I let now $\Gamma_{ij}^{ab} = \psi_{ab}(Z_i, Z_j, \gamma, \varphi, \alpha)$, where

$$\psi_{ab}(Z_i, Z_j, \gamma, \varphi, \alpha) = \varphi(a, X_i, b, X_j, \gamma_a, \gamma_b) + \phi_{ab}^{\gamma}(Z_i, Z_j, \gamma, \alpha^{\gamma}) + \phi_{ab}^{\varphi}(Z_i, Z_j, \varphi, \alpha^{\varphi}).$$

ψ_{ab} is the sum of an identifying function (φ_{ab}) and correction terms (($\phi_{ab}^{\gamma}, \phi_{ab}^{\varphi}$) needed to achieve orthogonality. For a given treatment rule π, orthogonal scores are given by

$$\Gamma_{ij}(\pi) = \sum_{(a,b)\in\{0,1\}^2} \psi_{ab}(Z_i, Z_j, \gamma, \varphi, \alpha)\pi_{ab}(X_i, X_j).$$

This framework accommodates SWFs in Section 3 if ψ_{ab} does not depend on Z_j and only depends on a so that it can be written as $\psi_a(Z_i, \gamma, \varphi, \alpha)$ for $a \in \{0, 1\}$. Hence, I only state conditions and results for SWFs in this Section. The intergenerational mobility example does not fit in the general setting, however, results extend to this example by Corollary 1 below. Next, I give conditions on the convergence of the nuisance estimators and the complexity of Π that allow me to prove asymptotic statistical guarantees for the estimator of treatment rules.

5.1 Conditions on the nuisance parameter estimators

I give high-level conditions for the estimators of nuisance parameters. The assumptions below are analogous to those in Escanciano and Terschuur (2023).

Assumption 4 $\mathbb{E}[|\psi(Z_i, Z_j, \gamma, \varphi, \alpha)|^2] < \infty$, *$\omega \in \{\gamma, \varphi\}$ and for* $(a, b) \in \{0, 1\}^2$

(i) $n^{\lambda_\gamma}\sqrt{\mathbb{E}(|\varphi(a, X_i, b, X_j, \hat{\gamma}_l) - \varphi(a, X_i, b, X_j, \gamma)|^2)} = o(1)$;

(ii) $n^{\lambda_\varphi}\sqrt{\mathbb{E}(|\hat{\varphi}_l(a, X_i, b, X_j, \gamma) - \varphi(a, X_i, b, X_j, \gamma)|^2)} = o(1)$;

(iii) $n^{\lambda_\gamma}\sqrt{\mathbb{E}(|\phi_{ab}^{\gamma}(Z_i, Z_j, \hat{\gamma}_l, \alpha^{\gamma}) - \phi_{ab}^{\gamma}(Z_i, Z_j, \gamma, \alpha^{\gamma})|^2)} = o(1)$;

(iv) $n^{\lambda_\varphi}\sqrt{\mathbb{E}(|\phi_{ab}^{\varphi}(Z_i, Z_j, \hat{\varphi}_l, \alpha^{\varphi}) - \phi_{ab}^{\varphi}(Z_i, Z_j, \varphi, \alpha^{\varphi})|^2)} = o(1)$;

(v) $n^{\lambda_\alpha}\sqrt{\mathbb{E}(|\phi_{ab}^{\omega}(Z_i, Z_j, \omega, \hat{\alpha}_l^{\omega}) - \phi_{ab}^{\omega}(Z_i, Z_j, \omega, \alpha^{\omega})|^2)} = o(1)$,

where $1/4 < \lambda_\gamma, \lambda_\varphi, \lambda_\alpha$.

These are mean-square consistency conditions for $\hat{\gamma}_l$, $\hat{\varphi}_l$ and $\hat{\alpha}_l$ separately. Assumption 4 often follows from the L2 convergence rates of the nuisance estimators. There is a large literature checking L_2-convergence rates for different machine learners under low-level sparsity or smoothness conditions on the nuisance parameters. The nonparametric literature gives rates for kernel regression and sieves/series (e.g. Chen (2007)). For L_1-penalty estimators such as Lasso see Belloni and Chernozhukov (2011) and Belloni and Chernozhukov (2013). For low-level conditions for shrinkage and kernel estimators see Appendix B in Sasaki and Ura (2021). Rates for L_2-boosting in low dimensions are found in Zhang and Yu (2005), and more recently Kueck et al. (2023) find rates for L_2-boosting with high dimensional data. For results on versions of random forests see Wager and Walther (2015) and Athey et al. (2019). For single-layer, sigmoid-based neural networks see Chen and White (1999) and for a modern setting of deep neural networks see Farrell et al. (2021). Note that $\hat{\varphi}_l$ estimates conditional expectations where both the dependent variables and the conditioning ones are indexed by both i and j. Stute (1991) calls such objects conditional U-statistics and studies the asymptotic properties of Nadaraya-Watson nonparametric estimators of these quantities. I run the machine learning algorithms on the stacked pairs. Unfortunately, not much is known about rates for such machine learning regressions which are computationally demanding. Define now the following interaction terms for $\omega \in \{\gamma, \varphi\}$ and let $||\cdot||$ denote the L2 norm.

$$
\begin{aligned}
\hat{\xi}_{ij,ab,l} &= \hat{\varphi}_l(a, X_i, b, X_j, \hat{\gamma}_l) - \varphi(a, X_i, b, X_j, \hat{\gamma}_l) - \hat{\varphi}_l(a, X_i, b, X_j, \gamma) + \varphi(a, X_i, b, X_j, \gamma), \\
\hat{\xi}^{\omega}_{ij,ab,l} &= \phi_{ab}(z_i, z_j, \hat{\omega}_l, \hat{\alpha}^{\omega}_l) - \phi_{ab}(z_i, z_j, \omega, \hat{\alpha}^{\omega}_l) - \phi_{ab}(z_i, z_j, \hat{\omega}_l, \alpha^{\omega}) + \phi_{ab}(z_i, z_j, \omega, \alpha^{\omega}).
\end{aligned}
$$

Assumption 5 *For each $l = 1, ..., L$*

(i) $\int\int \phi^{\gamma}_{ab}(z_i, z_j, \gamma, \hat{\alpha}^{\gamma}_l)F(dz_i)F(dz_j) = 0$ *and* $\int\int \phi^{\varphi}_{ab}(Z_i, Z_j, \varphi, \hat{\alpha}^{\varphi}_l)F(dz_i)F(dz_j) = 0$.

(ii) $\mathbb{E}(||\hat{\gamma}_l - \gamma||^2) = o(n^{-2\lambda_\gamma})$, $\mathbb{E}(||\hat{\varphi}_l - \varphi||^2) = o(n^{-2\lambda_\varphi})$ *and*

$$
\begin{aligned}
|\mathbb{E}[(\varphi(a, X_i, b, X_j, \tilde{\gamma}) + \phi^{\gamma}_{ab}(Z_i, Z_j, \tilde{\gamma}, \alpha^{\gamma}))\pi_{ab}(X_i, X_j)]| &\leq C||\tilde{\gamma} - \gamma||^2 \\
|\mathbb{E}[(\tilde{\varphi}(a, X_i, b, X_j, \gamma) + \phi^{\varphi}_{ab}(Z_i, Z_j, \tilde{\varphi}, \alpha^{\varphi}))\pi_{ab}(X_i, X_j)]| &\leq C||\tilde{\varphi} - \varphi||^2.
\end{aligned}
$$

Assumption 5 (i) is usually verified from visual inspection and (ii) requires L2 convergence rates and some smoothness. C is a constant so the right-hand-sides above do not depend on $\pi \in \Pi$.

Assumption 6 *For each $l = 1, ..., L$*

$$\sqrt{n}\mathbb{E}(\hat{\xi}^{\omega}_{ij,ab,l}\pi_{ab}(X_i, X_j)) = o(1).$$

These are rate conditions on the remainder terms $\hat{\xi}^{\omega}_{l}(w_i, w_j)$. Often, Assumption 6 follows if $\sqrt{n}||\hat{\alpha}^{\omega}_{l} - \alpha||||\hat{\omega}_{l} - \omega|| = o(1)$. This happens in Athey and Wager (2021).[2] This assumption allows to achieve $\sqrt{n}$ rates. In essence, it is enough for the product of the nonparametric estimators to go to zero at a $\sqrt{n}$ rate.

5.2 Conditions on the complexity of the policy class

The complexity of the policy class must also be restricted. If all sorts of subsets of $\mathcal{X}$ are allowed to decide who should be treated then we get overfitted policy rules. As in Athey and Wager (2021) I measure the policy class complexity with its VC dimension (see for instance Wainwright (2019)) which is allowed to grow with the sample size. Hence, from now on, I subscript the policy class by n, Π_n.

Assumption 7 *There are constants $0 < \beta < 1/2$ and $n^* \geq 1$ such that for all $n \geq n^*$, $VC(\Pi_n) < n^{\beta}$.*

Examples of finite VC-dimension classes are linear eligibility scores or generalized eligibility scores. Policy classes whose VC-dimension can increase with the sample size are for example decision trees which get deeper with sample size (see Athey and Wager (2021)).

5.3 Upper bounds

Let now

$$W(\pi) = \mathbb{E}\Big[\sum_{(a,b)\in\{0,1\}^2} \psi_{ab}(Z_i, Z_j, \gamma, \varphi, \alpha)\pi_{ab}(X_i, X_j)\Big],$$

$$\widetilde{W}_n(\pi) = \binom{n}{2}^{-1}\sum_{i<j}\Big[\sum_{(a,b)\in\{0,1\}^2} \psi_{ab}(Z_i, Z_j, \gamma, \varphi, \alpha)\pi_{ab}(X_i, X_j)\Big],$$

$$\hat{W}_n(\pi) = \binom{n}{2}^{-1}\sum_{l=1}^{L}\sum_{(i,j)\in I_l}\Big[\sum_{(a,b)\in\pi} \psi_{ab}(Z_i, Z_j, \hat{\gamma}_l, \hat{\varphi}_l, \hat{\alpha}_l)\pi_{ab}(X_i, X_j).\Big],$$

[2] In the proof of Lemma 4, invoking Cauchy-Schwarz inequality they get a bound on their interaction term that only depends on the product of L2 norms

$W(\pi)$ and $\widetilde{W}_n(\pi)$ are the welfare and the infeasible estimator of the welfare at policy rule π when all nuisance parameters are known. $\hat{W}_n(\pi)$ is the feasible estimator which we already introduced in (4.2). Let $W^*_{\Pi_n} = \sup_{\pi\in\Pi_n} W(\pi)$ be the best possible welfare. I give upper bounds to the regret: $\mathbb{E}[W^*_{\Pi_n} - W(\hat{\pi})]$. I start bounding the regret as follows

$$\begin{aligned}\mathbb{E}[W^*_{\Pi_n} - W(\hat{\pi})] &\leq 2\mathbb{E}\bigg[\sup_{\pi\in\Pi_n} |\hat{W}_n(\pi) - W(\pi)|\bigg] \\ &\leq 2\mathbb{E}\bigg[\sup_{\pi\in\Pi_n} |\hat{W}_n(\pi) - \widetilde{W}_n(\pi)|\bigg] + 2\mathbb{E}\bigg[\sup_{\pi\in\Pi_n} |\widetilde{W}_n(\pi) - W(\pi)|\bigg], \quad (5.1)\end{aligned}$$

where in the second inequality I have added and subtracted $\widetilde{W}_n(\pi)$ and used the triangle inequality. The second term above is a standard centered U-process indexed by $\pi \in \Pi_n$. I start as in Athey and Wager (2021) by showing the rate of convergence of this second term. I work for some fixed $(a, b) \in \{0, 1\}^2$ and let $\Pi_{ab,n} = \{\pi_{ab} : \pi \in \Pi_n\}$. The first step is to bound it by the Rademacher complexity which I define as

$$\mathcal{R}_n(\Pi_{ab,n}) = \mathbb{E}_\varepsilon\Bigg(\sup_{\pi\in\Pi_n}\Bigg|\lfloor n/2\rfloor^{-1}\sum_{i=1}^{\lfloor n/2\rfloor}\varepsilon_i\Gamma^{ab}_{i,\lfloor n/2\rfloor+i}\pi_{ab}(X_i, X_{\lfloor n/2\rfloor+i})\Bigg|\Bigg),$$

where F_ε is the distribution of Rademacher random variables taking value 1 and -1 with equal probability. $(\varepsilon_1, ..., \varepsilon_{\lfloor n/2\rfloor})$ are independent draws from F_ε. The next result gives a bound for the Rademacher complexity. Its result relies on a characterization of U-statistics as dependent sums of independent sums introduced in Hoeffding (1963) and greatly simplifies the proof by avoiding U-process theory.

Lemma 2

$$\mathbb{E}\bigg[\sup_{\pi\in\Pi_n} |\widetilde{W}_n(\pi) - W(\pi)|\bigg] \leq \mathbb{E}[2\mathcal{R}_n(\Pi_{ab,n})].$$

We want an asymptotic upper bound for $\mathbb{E}[\mathcal{R}_n(\Pi_{ab})]$. While Kitagawa and Tetenov (2018) and others provide bounds in terms of the maximum of the (bounded) scores, Athey and Wager (2021) provide bounds based on the variance. The next result provides a bound on the Rademacher complexity based on $S_{ab} = \mathbb{E}[\Gamma^{2\,ab}_{i,j}]$.

Lemma 3 *If Γ^{ab}_{ij} has bounded support for $(a, b) \in \{0, 1\}^2$, then under Assumptions 4 and 6*

$$\mathbb{E}[\mathcal{R}_n(\Pi_{ab,n})] = \mathcal{O}\Bigg(\sqrt{\frac{S_{ab}\cdot VC(\Pi_{ab,n})}{\lfloor n/2\rfloor}}\Bigg).$$

Lemma 3 can be generalized to sub-Gaussianity. However, it comes at the cost of making the (already involved) proofs substantially less tractable. Now I provide asymptotic upper bounds for the first term in (5.1). Escanciano and Terschuur (2023) show that for given $\pi \in \Pi_n$

$$\sqrt{n}(\hat{W}_n(\pi) - \widetilde{W}_n(\pi)) \to_p 0.$$

The next result makes the above uniform in $\pi \in \Pi_n$.

Lemma 4 (Uniform coupling) *Under Assumptions 4 and 6*

$$\sqrt{n}\mathbb{E}[\sup_{\pi \in \Pi_n} |\hat{W}_n(\pi) - \widetilde{W}_n(\pi)|] = \mathcal{O}\left(1 + \frac{VC(\Pi_{ab,n})}{\lfloor n/2 \rfloor^{\min(\lambda_\gamma, \lambda_\varphi, \lambda_\alpha)}}\right).$$

Finally, using Lemmas 3 and 4 the following holds.

Theorem 1 *If Assumptions 4, 6 and 7 hold with $\beta < \min(\lambda_\gamma, \lambda_\varphi, \lambda_\alpha)$. Then*

$$\mathbb{E}[W^*_{\Pi_n} - W(\hat{\pi})] = \mathcal{O}\left(\sqrt{\frac{S_{ab} \cdot (2VC(\Pi_n) - 1)}{\lfloor n/2 \rfloor}}\right).$$

Corollary 1 *The bound in Theorem 1 applies to the Intergenerational mobility example.*

5.4 Computational aspects

The U-statistic nature of the welfare functionals analyzed in this paper inevitably increases the computational complexity of welfare estimation for a fixed rule from $O(n)$ to $O(n^2)$. An interesting idea for further research would be to reduce this computational burden by the use of incomplete U-statistics which only sum across a much smaller sample of the $\binom{n}{2}$ pairs. Dependence among pairs allows omitting many summands retaining desirable first order properties of U-statistics (see Lee (2019)). It is likely that similar results could be obtained in this policy learning setting.

The computational burden of scanning through the policy class can also be affected by the U-statistic nature of the problem. For instance, when building trees, observation i and j might be in terminal nodes of different sub-trees. This means that splits in a given sub-tree depend on other sub-trees. Hence, we cannot use the computational shortcuts in Athey and Wager (2021) where each sub-tree can be optimized independently.

In the case of linear rules $\pi_\beta(X_i) = 1(X_i'\beta \geq 0)$ for $\beta \in B \subset \mathbb{R}^{k+1}$ (with the intercept included in X_i), I show in Appendix C that we can recast the problem as a Mixed Integer Linear Program (MILP) as in Kitagawa and Tetenov (2018). This formulation adds $O(n^2)$ linear constraints to a continuous choice variable but, importantly, keeps the number of binary choice variables at order $O(n)$.

A final avenue for future research is to frame the U-statistic policy learning problem as some sort of weighted classification/ranking problem in the same way that the usual linear policy learning problem can be framed as a weighted classification problem (see Kitagawa et al. (2021)). In these contexts, successful algorithms have been developed that replace the objective function by a convex function and use some form of stochastic gradient descent to find approximate solutions using parametric, support vector or neural network based policies. However, the pairwise nature of the objective combined with the individualized nature of the treatment rule makes the adoption of this approach less straightforward.

Also, the approach of convex surrogates works well whenever we are finding the optimal policy within a complex class which we can argue contains the unconstrained optimum (e.g. dense neural networks). In policy learning, it is often the case that we want to restrict the policy class due to interpretability or legal constraints. In these cases, the analysis of convex surrogates becomes much more challenging, see Kitagawa et al. (2021).

6 Empirical illustration

I study the optimal allocation of children to preschool. I make use of the Panel Study of Income Dynamics (PSID) database which has been following families for nearly 50 years. The nature of this survey allows us to observe a rich set of circumstances and long-term outcomes. In 1995, PSID asked adults between 18-30 years old about their participation in preschool. Hence, we can track the long-term outcomes of these individuals. I take as an outcome the average earnings from 25 to 35 years old. I assume selection on sex, birthyear, average parental income in the 5 years before birth, mother's education, father's education, father's occupation and whether the individual is black holds. In Table 1 we see the results of estimating the Average Treatment Effect (ATE), Gini, IOp and Intergenerational mobility measured by the Kendall-τ.

To estimate regressions and propensity scores I use Conditional Inference Forests

(CIFs) from Hothorn et al. (2006) implemented in the R package `party`. CIFs are an ensemble of decision trees where variable selection and splitting are done via statistical tests. They have become popular in the IOp literaure (e.g. Brunori et al. (2021)). I choose CIF by cross-validation among a pool of different machine learners.

Outcome	ATE	se	p-value	Gini	IOp	IGM	n
Earnings 25-35	4622.063	1083.865	0	0.392	0.172	0.168	2971

Table 1: ATE, Gini, IOp and Kendal-τ

To estimate the ATE, I use the doubly robust Augmented Inverse Propensity weighting from Robins et al. (1994) using Conditional Inference Forests (CIF) to estimate the regression functions and propensity scores. Under the assumption of no selection on observables, we observe a positive effect of attending preschool of 4,622$ of added annual earnings. Dollars have been adjusted by the CPI to 2010 dollars. The Gini coefficient is 0.39 and IOp is 0.17, i.e. almost 44% of total inequality can be explained by the circumstances we observe. The Kendall-τ is around 0.17 which indicates a positive association between parental and child incomes.

I compute optimal treatment rules based on parental income and the mother's years of education. I set the target in the Kendall-τ welfare to zero, meaning that the aim is to completely erase intergenerational persistence. As the policy class, I use 2-depth decision trees. I use the sample deciles of parental income as cutting points instead of all the observed values of parental income. When computing optimal rules I scan across all possible 2-depth decision trees. Hence, the search exhausts all possible treatment rules in this class.

Optimal treatment allocation is the same for additive and IOp welfare. Although this seems surprising, it is possible if decreasing inequality of opportunity leads to sizeable reductions of the average. In fact, as reported in Table 2, the estimated rule maximizing the average already drastically decreases IOp. I show the optimal rule under these two welfares in Figure 1. At the terminal nodes, I report the number of observations, the conditional average treatment effect (CATE) in the node and the proportion of observations node that are treated in the data ($\hat{p}$). For additive/IOp welfare, the first cutting point is whether parental income is below or above the 40th percentile (51,515$). If an observation is below this cut-off the tree splits according to the education of the mother. If parental income is below the 40th percentile and the mother's education is less than college (below 13 years) the tree allocates the observation to treatment. The CATE in this node is positive so, as we would expect, an additive policy maker treats these observations. If parental income is below the 40th percentile but the mother is highly educated the CATE is negative and hence the additive policy maker does not allocate the individual to treatment. For high parental income, we split next on the mother's education but at a higher level. If your parental income is higher than the 40th percentile and your mother attended college or less (16 years of education or less) you are allocated to treatment and in this node, we have large positive effects. However, we do not allocate kids with high parental income and high maternal education to preschool since the CATE in this group is negative.

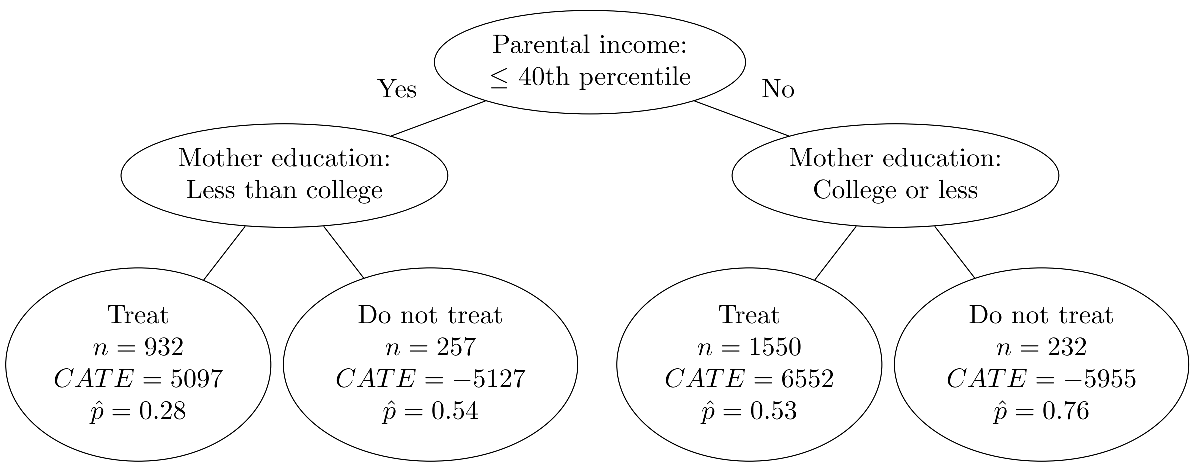

Figure 1: Estimated optimal policy rule under additive and IOp welfare.

I conjecture that there is a substitution effect between the quality of the preschool institution and mother's education. If we take parental income to be a proxy for the quality of preschool, it is enough for the mother to have more than 13 years of education for the child to be better off without preschool. However, for children who attend better preschools (have higher parental income), the mother has to have more than 16 years of education for the child to be better off without preschool. This is in line with results in the psychology and economics literature which document that in early education institutions, there is less interaction with adults and hence there can be a negative effect of attending such institutions if the interactions with adults in the household are of "high-quality" (see Fort et al. (2020)). To decrease IOp further, we need to treat advantaged kids who do not benefit from preschool. The penalization of IOp is not severe enough to do this. The treatment allocation observed in the data is quite far from the optimal rule. This is sensible since future earnings are not the only consideration when parents decide whether to send their kids to preschool.

In figure 2, we see the optimal policy rule for the inequality SWF, i.e. now we penalize all inequalities and not just the ones explained by circumstances. The tree is the same except for the first cutting point on parental income. We first divide individuals into those with parental income lower and higher than the 20th percentile (37,699$). Then, the division based on the mother's education is the same. Hence, compared to the previous tree, we shift 20% of the population to the right side subtree. For instance, a kid who has a parental income of 40,000$ and whose mother has 16 years of education is not treated under the additive/IOp welfare but is treated under the inequality based optimal rule. Although masked by other observations in the node, this 20% of the population who is switched to treatment has an estimated negative CATE. When we penalize all sorts of inequalities, it starts becoming optimal to decrease the average outcome to decrease inequality.

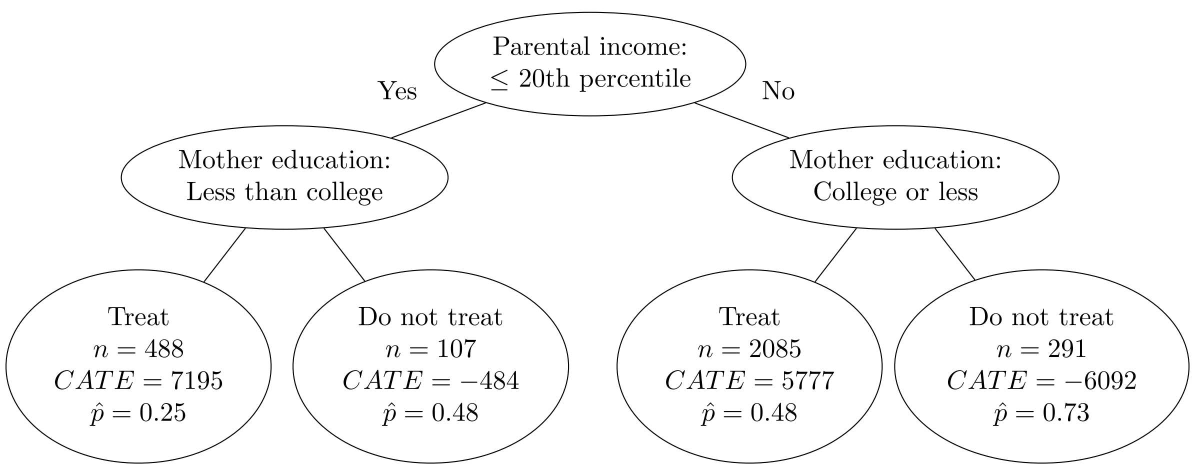

Figure 2: Estimated optimal policy rule under inequality welfare.

In Figure 3 we see the results for an intergenerational mobility aware SWF. Notice that in the intergenerational mobility welfare there is no efficiency motive and we target a zero Kendall-τ. Hence, the optimal policy is even more controversial since individuals with positive treatment effects are not treated and individuals with negative treatment effects are treated.If the mother has more than 13 years of education but less than 16 years of education (college education), the optimal policy does not treat even though there are positive treatment effects. Even more extreme, if the mother is very highly educated, the optimal policy treats even though there are negative treatment effects. For kids with maternal education below college, the optimal policy treats only if you are in the lowest 50% of the distribution of parental income (below 58,270$). If your mother has less than college but your parents are in the richest half of the income distribution you are not treated even though there are positive treatment effects.

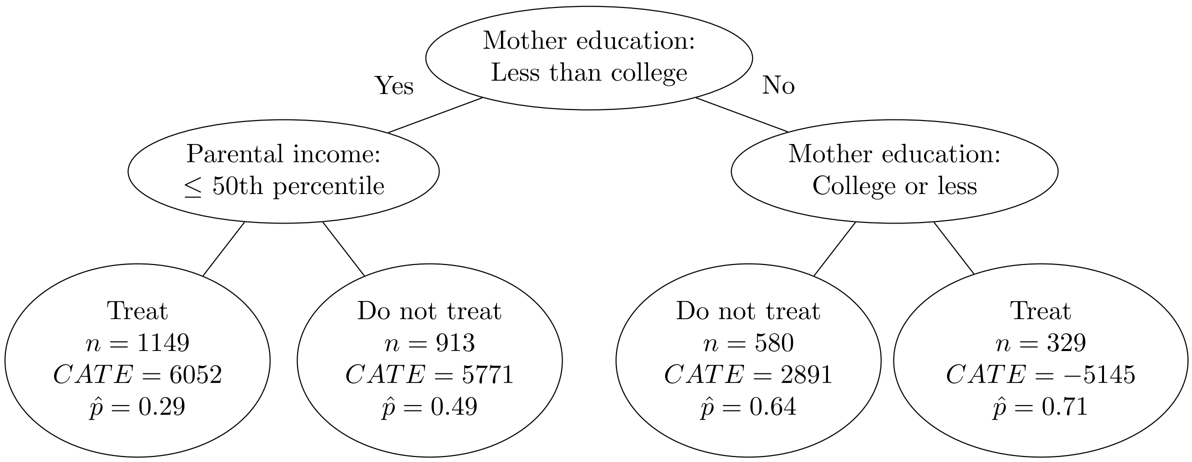

Figure 3: Estimated optimal policy rule under intergenerational mobility welfare.

Finally, in Table 2 we see a summary of the results and compare the estimated optimal treatments with situations in which no one or everyone is treated. For additive, IOp and inequality welfares, treating no one gives the worst welfare. In the IGM case, treating no one and treating everyone give basically the same welfare (note that maximal welfare in the IGM case is 0). The additive and IOp welfares have an estimated optimal policy rule that attains the highest average outcome and the lowest IOp. The decrease in IOp under this rule is drastic. While in the sample we can explain 44% of total inequality with circumstances (IOp/Gini), at the optimal additive/IOp rule we explain 35%. This explains why both rules coincide. In other settings, maximizing the average might increase IOp.

The estimated optimal treatment rule for IGM gives a similar IOp as the the estimated optimal policy rule for IOp welfare, but at a much larger cost in average outcome. The estimated optimal policy rule under the inequality welfare gives the lowest Gini compared to additive and IOp welfares. Interestingly, it gives the highest IOp across all welfares. The IGM welfare estimated rule gives virtually the same Gini as the inequality welfare but at a much higher cost in average outcome. IGM rule gives the lowest Kendall-τ.

Finally, comparing the results with what we observe with the treatment allocation in the sample, we achieve a higher mean with all other welfares except with the IGM welfare. The Gini in the sample is the same as the one under the estimated optimal additive and IOp rule. The observed IOp in the sample is higher than the one achieved

		Welfare	Mean	Gini	IOp	IGM	Share treated
Additive	**Optimal rule**	39727	39727	0.392	0.138	0.15	0.84
	Treat no one	34169	34169	0.4	0.162	0.148	0
	Treat everyone	38778	38778	0.383	0.142	0.15	1
IOp	**Optimal rule**	34231	·	·	·	·	·
	Treat no one	28640	·	·	·	·	·
	Treat everyone	33282	·	·	·	·	·
Inequality	**Optimal rule**	24165	39383	0.386	0.141	0.153	0.87
	Treat no one	20518	34169	0.4	0.162	0.148	0
	Treat everyone	23942	38778	0.383	0.142	0.15	1
IGM	**Optimal rule**	-0.086	35951	0.383	0.139	0.086	0.5
	Treat no one	-0.148	34169	0.4	0.162	0.148	0
	Treat everyone	-0.15	38778	0.383	0.142	0.15	1
Sample	·	·	36197	0.392	0.172	0.168	0.47

Table 2: Welfare, mean, Gini, IOp, IGM and share treated for different optimal policy rules compared with policies which treat no one and everyone. I also show the actual values observed in the sample. The dots in the IOp rows indicate that the optimal policy rule is the same as in the additive case.

under the estimated rules of all other welfares. IGM observed in the sample is the lowest (highest Kendall-τ) compared to all welfares. Finally, the share of treated in the sample is also lower than the one achieved under the estimated optimal rule of all other welfares. However, this could be explained by not taking into account costs of treatment.

7 Conclusion

This paper extends previous work on policy learning to accommodate general semi-parametric SWFs estimable by U-statistics. This opens the analysis to highly policy-relevant SWFs such as inequality, inequality of opportunity and intergenerational mobility aware SWFs. The inequality of opportunity SWF is especially useful when we do not want to penalize all sorts of inequality but just unfair sources of inequality. In the empirical illustration, inequality and IGM aware SWFs assign groups with negative treatment effects to treatment. However, the additive and IOp rules coincide. Further work is needed, particularly to ease the computational burden of the method. The application of convex surrogates in Kitagawa et al. (2021) is a promising avenue to achieve this. Another interesting extension is to allow for multiple treatments as in Zhou et al. (2023) in a U-statistics setting. In our illustration, this could be useful to study the optimal allocation of children to different types of preschools. Finally, it would be interesting to extend the results to the case of continuous treatments as in Athey and Wager (2021).

8 Appendix

A Auxiliary lemmas

Now some lemmas needed for the main results. For a fixed $\{X_i, \Gamma_{i,\lfloor n/2\rfloor+i}\}_{i=1}^{\lfloor n/2\rfloor}$ define

$$\widetilde{\Pi}_{ab} = \{\pi_{ab}(X_1, X_{\lfloor n/2\rfloor+1}), ..., \pi_{ab}(X_{\lfloor n/2\rfloor}, X_n) : \pi \in \Pi\}.$$

For $\pi, \pi' \in \widetilde{\Pi}_{ab}$ define the following distances

$$D_n^2(\pi, \pi') = \frac{\sum_{i=1}^{\lfloor n/2\rfloor} \Gamma_{i,\lfloor n/2\rfloor+i}^{2\,ab}(\pi_{ab}(X_i, X_{\lfloor n/2\rfloor+i}) - \pi'_{ab}(X_i, X_{\lfloor n/2\rfloor+i}))^2}{\sum_{i=1}^{\lfloor n/2\rfloor} \Gamma_{i,\lfloor n/2\rfloor+i}^{2\,ab}},$$

$$H(\pi, \pi') = \frac{1}{n}\sum_{i=1}^{n} \mathbb{1}(\pi_{ab}(X_i, X_{\lfloor n/2\rfloor+i}) \neq \pi'_{ab}(X_i, X_{\lfloor n/2\rfloor+i})).$$

Let $N_{D_n}(\varepsilon, \widetilde{\Pi}_{ab}, \{X_i, \Gamma_{i,\lfloor n/2\rfloor+i}\}_{i=1}^{\lfloor n/2\rfloor})$ be the number of balls of radius ε needed to cover $\widetilde{\Pi}_{ab}$ under distance D_n. Define the same object for the Hamming distance H and let

$$N_H(\varepsilon, \widetilde{\Pi}_{ab}) = \sup\{N_H(\varepsilon, \widetilde{\Pi}_{ab}, \{X_i\}_{i=1}^m) : X_1, ..., X_m \in \mathcal{X}, m \geq 1\}.$$

Note $N_H(\varepsilon, \widetilde{\Pi}_{ab})$ does not depend on m. It will be useful to bound N_{D_n} with N_H.

Lemma 5 *For fixed* $\{X_i, \Gamma_{i,\lfloor n/2\rfloor+i}\}_{i=1}^{\lfloor n/2\rfloor}$ *we have that* $N_{D_n}(\varepsilon, \widetilde{\Pi}_{ab}, \{X_i, \Gamma_{i,\lfloor n/2\rfloor+i}\}_{i=1}^{\lfloor n/2\rfloor}) \leq N_H(\varepsilon^2, \widetilde{\Pi}_{ab})$.

Proof: Take an auxiliary sample $\{X'_j\}_{j=1}^m$ contained in $\{X_i\}_{i=1}^n$ such that

$$\left| |B_i| - \frac{m\Gamma_{i,\lfloor n/2\rfloor+i}^{2\,ab}}{\sum_{i=1}^{\lfloor n\rfloor/2} \Gamma_{i,\lfloor n/2\rfloor+i}^{2\,ab}} \right| \leq 1,$$

where $B_i = \{j \in \{1, ..., m\} : X'_j = X_i\}$. Then, for $\pi, \pi' \in \widetilde{\Pi}_{ab}$

$$D_n^2(\pi, \pi') = \frac{1}{m}\sum_{i=1}^{\lfloor n/2\rfloor} \underbrace{\frac{m\Gamma_{i,\lfloor n/2\rfloor+i}^{2\,ab}}{\sum_{k=1}^{\lfloor n/2\rfloor} \Gamma_{k,\lfloor n/2\rfloor+k}^{2\,ab}}}_{\geq |B_i|-1} \mathbb{1}(\pi_{ab}(X_i, X_{\lfloor n/2\rfloor+i}) \neq \pi'_{ab}(X_i, X_{\lfloor n/2\rfloor+i})).$$

So

$$\begin{aligned}
D_n^2(\pi,\pi') &\geq \sum_{i=1}^{\lfloor n/2\rfloor} \frac{|B_i|}{m} \mathbb{1}(\pi_{ab}(X_i, X_{\lfloor n/2\rfloor+i}) \neq \pi'_{ab}(X_i, X_{\lfloor n/2\rfloor+i})) - O(1/m) \\
&= \sum_{i=1}^{\lfloor n/2\rfloor} \frac{|B_i|}{m} \frac{1}{|B_i|} \sum_{j\in B_i} \mathbb{1}(\pi_{ab}(X'_j, X'_{\lfloor n/2\rfloor+j}) \neq \pi'_{ab}(X'_j, X'_{\lfloor n/2\rfloor+j})) - O(1/m) \\
&= \frac{1}{m} \sum_{i=1}^{\lfloor n/2\rfloor} \sum_{j\in B_i} \mathbb{1}(\pi_{ab}(X'_j, X'_{\lfloor n/2\rfloor+j}) \neq \pi'_{ab}(X'_j, X'_{\lfloor n/2\rfloor+j})) - O(1/m).
\end{aligned}$$

The first equality uses that all summands in the inner sum are the same since for all $j \in B_i$ we know that $(X_i, X_{\lfloor n/2\rfloor+i}) = (X'_j, X'_{\lfloor n/2\rfloor+j})$. The sum $\sum_{i=1}^{\lfloor n/2\rfloor} \sum_{j\in B_i}$ can sum pairs more than once (e.g. if $(X_1, X_{\lfloor n/2\rfloor+1}) = (X_2, X_{\lfloor n/2\rfloor+2})$ then $B_1 = B_2$). Since $\{X'_j\}_{j=1}^m$ is contained in $\{X'_i\}_{i=1}^m$

$$\begin{aligned}
D_n^2(\pi,\pi') &\geq \frac{1}{m} \sum_{j=1}^{m} \mathbb{1}(\pi_{ab}(X'_j, X'_{\lfloor n/2\rfloor+j}) \neq \pi'_{ab}(X'_j, X'_{\lfloor n/2\rfloor+j})) - O(1/m) \\
&= H(\pi,\pi') - O(1/m).
\end{aligned}$$

So, $H(\pi,\pi') \leq D_n^2(\pi,\pi') + O(1/m)$. N_H does not depend on m, so making m large we conclude

$$N_{D_n}(\varepsilon, \widetilde{\Pi}_{ab}, \{X_i, \Gamma_{i,\lfloor n/2\rfloor+i}\}_{i=1}^{\lfloor n/2\rfloor}) \leq N_H(\varepsilon^2, \widetilde{\Pi}_{ab}).$$

■

Now we prove that the sequence of covers we use in the proof of Lemma 3 exists.

Lemma 6 *$\exists$ $\{B_k\}_{k=0}^K$ with $K < \infty$ of $\widetilde{\Pi}_{ab}$ with $B_k \subset \widetilde{\Pi}_{ab}$ s.t. for $k = 0, ..., K$*

- *For all $\pi \in \widetilde{\Pi}_{ab}$, there exists $b \in B_k$ such that $D_n(\pi, b) \leq 2^{-k}$,*
- $|B_k| = N_{D_n}(2^{-k}, \widetilde{\Pi}_{ab}, \{X_i, \Gamma_{i,\lfloor n/2\rfloor+i}\}_{i=1}^{\lfloor n/2\rfloor}) \leq |\widetilde{\Pi}_{ab}|$.

Proof: Note that $|\widetilde{\Pi}_{ab}| < 2^{\lfloor n/2\rfloor} < \infty$ since X_i's are fixed. Since $\widetilde{\pi}_{ab}$ is finite and $B_k \subset \widetilde{\Pi}_{ab}$ for all k, there is finite K s.t. we can set $B_K = \widetilde{\Pi}_{ab}$. This is because for any B_k which is a strict subset of $\widetilde{\Pi}_{ab}$ there is $\pi \in \widetilde{\Pi}_{ab}$ s.t for all $b \in B_k$, $D_n(b,\pi) > a > 0$ and there is $K > 0$ s.t. $2^{-K} < a$. K is finite since there are only finitely many subsets of $\widetilde{\Pi}_{ab}$. For B_{K-1} we look through all possible strict subsets for one which satisfies our conditions, if there is none we know that $B_{K-1} = \widetilde{\Pi}_{ab}$ does satisfy them. In this way, we can go backwards and build the sequence of covers. ■

Lemma 7 $VC(\widetilde{\Pi}_{ab}) \leq 2VC(\Pi) - 1.$

Proof: Let $\pi_t(X_i) = \mathbb{1}(\pi(X_i) = t)$ for $t \in \{0, 1\}$. Define $\Pi_t = \{\mathbb{1}(\pi(X_i) = t) : \pi \in \Pi\}$. Note that $\Pi_1 = \Pi$ and that $VC(\Pi_0) = VC(\Pi_1)$ by Lemma 9.7 in Kosorok (2008). Now note that for any $(a, b) \in \{0, 1\}^2$

$$\widetilde{\Pi}_{ab} = \{\pi_a \cdot \pi_b : (\pi_a, \pi_b) \in \Pi_a \times \Pi_b\},$$

so Lemma 9.9 (ii) in Kosorok (2008) yields the desired result. ■

B Proofs of main results

Proof of Proposition 3.1: I proof only the identification of the first term of the welfare since the second one follows in the same manner.

$$\begin{aligned}
\mathbb{E}[g(Y(1), X, \gamma^{(1)})\pi(X)] &= \mathbb{E}[\mathbb{E}(g(Y(1), X, \gamma(1, X))|X)\pi(X)] \\
&= \mathbb{E}[\mathbb{E}(g(Y, X, \gamma(1, X)|D_i = 1, X_i)\pi(X_i)] \\
&= \mathbb{E}[\varphi(1, X, \gamma)\pi(X_i)].
\end{aligned}$$

■

Proof of Lemma 1:

$$\begin{aligned}
\frac{d}{d\tau}\mathbb{E}[\varphi(1, X_i, \gamma_\tau(1, X_i))\pi(X_i)] &= \frac{d}{d\tau}\mathbb{E}[\mathbb{E}[g(Y_i, X_i, \gamma_\tau(1, X_i))\pi(X_i)|D_i = 1, X_i]] \\
&= \frac{d}{d\tau}\mathbb{E}\left[\mathbb{E}\left[\frac{g(Y_i, X_i, \gamma_\tau(D, X_i))D}{e(X_i)}\pi(X_i)|X_i\right]\right] \\
&= \frac{d}{d\tau}\mathbb{E}\left[\frac{g(Y_i, X_i, \gamma_\tau(D, X_i))D}{e(X_i)}\pi(X_i)\right] \\
&= \frac{d}{d\tau}\mathbb{E}\left[\alpha_1(D_i, X_i)\gamma_\tau(D_i, X_i)\pi(X_i)\right],
\end{aligned}$$

where we use Assumption 2 for the last equality. Same arguments hold for the $D_i = 0$ case. ■

Proof of Proposition 3.2: Let $d/d\tau$ be the derivative with respect to τ evaluated at $\tau = 0$, let $\varphi_\tau = \varphi + \tau\tilde{\varphi}$ for some $\tilde{\varphi}$ in the space where φ lives and $\mathbb{E}_\tau$ be the expectation with respect to $F + \tau(H - F)$ for some alternative distribution H. Then

$$\frac{d}{d\tau}\mathbb{E}[\varphi_\tau(1, X_i, \gamma_\tau(1, X_i))\pi(X_i)] = \frac{d}{d\tau}\mathbb{E}[\varphi_\tau(1, X_i, \gamma(1, X_i))\pi(X_i)] + \frac{d}{d\tau}\mathbb{E}[\varphi(1, X_i, \gamma_\tau(1, X_i))\pi(X_i)].$$

For the first term note that

$$\begin{aligned}
\frac{d}{d\tau}\mathbb{E}[\varphi_\tau(1,X_i,\gamma(1,X_i))\pi(X_i)] &= \frac{d}{d\tau}\mathbb{E}\left[\frac{D_i}{e(X_i)}\varphi_\tau(1,X_i,\gamma(1,X_i))\pi(X_i)\right] \\
&= \frac{d}{d\tau}\mathbb{E}\left[\frac{D_i}{e(X_i)}\varphi_\tau(D_i,X_i,\gamma(D_i,X_i))\pi(X_i)\right] \\
&= \frac{d}{d\tau}\mathbb{E}_\tau\left[\frac{D_i}{e(X_i)}\varphi_\tau(D_i,X_i,\gamma(D_i,X_i))\pi(X_i)\right] \\
&- \frac{d}{d\tau}\mathbb{E}_\tau\left[\frac{D_i}{e(X_i)}\varphi(D_i,X_i,\gamma(D_i,X_i))\pi(X_i)\right] \\
&= \frac{d}{d\tau}\mathbb{E}_\tau\left[\frac{D_i}{e(X_i)}(g(Y_i,X_i,\gamma(D_i,X_i)) - \varphi(D_i,X_i,\gamma(D_i,X_i)))\pi(X_i)\right],
\end{aligned}$$

using LIE in the first equality, that $\varphi_\tau(1,X_i,\gamma(1,X_i))D_i = \varphi_\tau(D_i,X_i,\gamma(D_i,X_i))D_i$ a.s., then the chain rule and that $\varphi_\tau(D_i,X_i,\gamma(D_i,X_i))$ is a projection of $g(Y_i,X_i,\gamma(D_i,X_i))$. For the second term, by Lemma 1 we have

$$\begin{aligned}
\frac{d}{d\tau}\mathbb{E}[\varphi(1,X_i,\gamma_\tau(1,X_i))\pi(X_i)] &= \frac{d}{d\tau}\mathbb{E}[\alpha_1(D_i,X_i)\gamma_\tau(D_i,X_i)\pi(X_i)] \\
&= \frac{d}{d\tau}\mathbb{E}_\tau[\alpha_1(D_i,X_i)\gamma_\tau(D_i,X_i)\pi(X_i)] \\
&- \frac{d}{d\tau}\mathbb{E}_\tau[\alpha_1(D_i,X_i)\gamma_\tau(D_i,X_i)\pi(X_i)] \\
&= \frac{d}{d\tau}\mathbb{E}_\tau[\alpha_1(D_i,X_i)(Y_i - \gamma(D_i,X_i))\pi(X_i)],
\end{aligned}$$

using the chain rule and then the fact that γ is a projection. Then, following Chernozhukov et al. (2022) we have that

$$\Gamma_{1i} = \varphi(1,X_i,\gamma) + \frac{D_i}{e(X_i)}(g(Y_i,X_i,\gamma) - \varphi(D_i,X_i,\gamma)) + \alpha_1(D_i,X_i)(Y_i - \gamma(D_i,X_i)).$$

The arguments for Γ_{0i} are the analogous. ■

Proof of Proposition 3.3:

$$
\begin{aligned}
\psi_j(\bar\varphi,\bar r,\gamma,\bar\alpha_j) &= \mathbb{E}\left[\bar\varphi(1,X,\gamma)\right] + \mathbb{E}\left[\bar r(D,X)\left(g(Y,X,\gamma(D,X)) - \bar\varphi(D,X,\gamma)\right)\right] \\
&+ \underbrace{\mathbb{E}\left[\bar\alpha_j(D,X)(Y-\gamma(D,X))\right]}_{=0} \\
&= \mathbb{E}\left[\bar\varphi(D,X,\gamma)r(D,X)\right] + \mathbb{E}\left[\bar r(D,X)\left(g(Y,X,\gamma(D,X)) - \bar\varphi(D,X,\gamma)\right)\right] \\
&- \underbrace{\mathbb{E}\left[\bar r(D,X)(g(Y,X,\gamma(D,X)) - \varphi(D,X,\gamma))\right]}_{=0} \\
&= \mathbb{E}\left[\bar\varphi(D,X,\gamma)r(D,X)\right] + \mathbb{E}\left[\bar r(D,X)\left(\varphi(D,X,\gamma) - \bar\varphi(D,X,\gamma)\right)\right] \\
&+ \psi_j(\varphi,r,\gamma,\alpha_j) - \mathbb{E}\left[\varphi(D,X,\gamma)r(D,X)\right] \\
&= \psi_j(\varphi,r,\gamma,\alpha_j) + \mathbb{E}\left[(\bar\varphi(D,X,\gamma)-\varphi(D,X,\gamma))(r(D,X)-\bar r(D,X))\right].
\end{aligned}
$$

If g is affine in its third argument, then

$$
\begin{aligned}
&\psi_j(\bar\varphi,\bar r,\bar\gamma,\bar\alpha_j) - \psi_j(\varphi,r,\gamma,\alpha_j) \\
&\qquad= \underbrace{\mathbb{E}\left[\bar\varphi(1,X,\bar\gamma)\right] + \mathbb{E}\left[\bar r(D,X)\left(g(Y,X,\bar\gamma(D,X)) - \bar\varphi(D,X,\bar\gamma)\right)\right] - \mathbb{E}[\varphi(1,X,\gamma)]}_{(1)} \\
&\qquad+ \underbrace{\mathbb{E}\left[\bar\alpha_j(D,X)(Y-\bar\gamma(D,X))\right] - \mathbb{E}[\alpha_j(D,X)(Y-\gamma(D,X))]}_{(2)}.
\end{aligned}
$$

I first address term (1). Note that

$$
\begin{aligned}
(1) &= \underbrace{\mathbb{E}\left[\bar\varphi(1,X,\bar\gamma)\right] + \mathbb{E}\left[\bar r(D,X)\left(g(Y,X,\bar\gamma(D,X)) - \bar\varphi(D,X,\bar\gamma)\right)\right] - \mathbb{E}[\varphi(1,X,\bar\gamma)]}_{(1.1)} \\
&+ \underbrace{\mathbb{E}[\varphi(1,X,\bar\gamma)] - \mathbb{E}[\varphi(1,X,\gamma)]}_{(1.2)}.
\end{aligned}
$$

I proceed with (1.1)

$$
\begin{aligned}
(1.1) &= \mathbb{E}\left[\bar\varphi(1,X,\bar\gamma) - \varphi(1,X,\bar\gamma)\right] \\
&+ \mathbb{E}\left[\bar r(D,X)\left(g(Y,X,\bar\gamma(D,X)) - \varphi(D,X,\bar\gamma) - \bar\varphi(D,X,\bar\gamma) + \varphi(D,X,\bar\gamma)\right)\right] \\
&= \mathbb{E}\left[r(D,X)\left(\bar\varphi(D,X,\bar\gamma) - \varphi(D,X,\bar\gamma)\right)\right] - \mathbb{E}\left[\bar r(D,X)\left(\bar\varphi(D,X,\bar\gamma) - \varphi(D,X,\bar\gamma)\right)\right] \\
&= \mathbb{E}\left[(r(D,X) - \bar r(D,X))\left(\bar\varphi(D,X,\bar\gamma) - \varphi(D,X,\bar\gamma)\right)\right],
\end{aligned}
$$

where in the second equality I have used that $\varphi(D,X,\bar\gamma)$ is a projection of $g(Y,X,\bar\gamma(D,X))$.

Since g is affine in its third argument, there is $\tilde{g}$ and z such that $g(Y, X, \bar{\gamma}(D, X)) = \tilde{g}(Y, X)\bar{\gamma}(D, X) + z(Y, X)$. Let $\xi(D, X) = \mathbb{E}[\tilde{g}(Y, X)|D, X]$. Then, addressing term (1.2) we have that by Assumption 2

$$\begin{aligned}
(1.2) &= \mathbb{E}[\xi(D, X)(\bar{\gamma}(D, X) - \gamma(D, X))] \\
&= \mathbb{E}[\alpha_j(D, X)(\bar{\gamma}(D, X) - \gamma(D, X))].
\end{aligned}$$

For (2)

$$\begin{aligned}
(2) &= \mathbb{E}[\bar{\alpha}(D, X)(Y - \bar{\gamma}(D, X))] \\
&= \mathbb{E}[\bar{\alpha}(D, X)(Y - \bar{\gamma}(D, X))] - \mathbb{E}[\bar{\alpha}(D, X)(Y - \gamma(D, X))] \\
&= -\mathbb{E}[\bar{\alpha}(D, X)(\bar{\gamma}(D, X) - \gamma(D, X))].
\end{aligned}$$

Hence,

$$\begin{aligned}
(1) + (2) &= \mathbb{E}[(r(D, X) - \bar{r}(D, X))(\bar{\varphi}(D, X, \bar{\gamma}) - \varphi(D, X, \bar{\gamma}))] \\
&\quad + \mathbb{E}[(\alpha_j(D, X) - \bar{\alpha}_j(D, X))(\bar{\gamma}(D, X) - \gamma(D, X))].
\end{aligned}$$

■

Proof of Proposition 4.1:

$$\begin{aligned}
W(\pi) &= \mathbb{E}\left[\sum_{(a,b)\in\{0,1\}^2} g(Y_i(a), X_i, Y_j(b), X_j, \gamma^{(a)}, \gamma^{(b)})\pi_{ab}(X_i, X_j)\right] \\
&= \mathbb{E}\left[\mathbb{E}\left(\sum_{(a,b)\in\{0,1\}^2} g(Y_i(a), X_i, Y_j(b), X_j, \gamma^{(a)}, \gamma^{(b)})\middle| X_i, X_j\right)\pi_{ab}(X_i, X_j)\right] \\
&= \mathbb{E}\left[\mathbb{E}\left(\sum_{(a,b)\in\{0,1\}^2} g(Y_i, X_i, Y_j, X_j, \gamma(a, X_i), \gamma(b, X_j))\middle| X_i, D_i = a, X_j, D_j = b\right)\pi_{ab}(X_i, X_j)\right].
\end{aligned}$$

where in the second equality I use LIE, in the third I use selection on observables. ■

Proof of Proposition 4.2: Let $d/d\tau$ be the derivative at $\tau = 0$. Let $\varphi_\tau = \varphi + \tau\tilde{\varphi}$ for some $\tilde{\varphi} \in \mathcal{G}$. By the chain rule

$$\frac{d}{d\tau}\mathbb{E}[\varphi_\tau(a, X_i, b, X_j, \gamma_\tau)] = \frac{d}{d\tau}\mathbb{E}[\varphi(a, X_i, b, X_j, \gamma_\tau)] + \frac{d}{d\tau}\mathbb{E}[\varphi_\tau(a, X_i, b, X_j, \gamma)].$$

By Assumption 3 we have that the first term is

$$\frac{d}{d\tau}\mathbb{E}[\varphi(a,X_i,b,X_j,\gamma_\tau)] = \frac{d}{d\tau}\mathbb{E}\left[\sum_{p=1}^{P}\alpha^{\gamma}_{ab,p}(D_i,X_i,D_j,X_j)(c_{1p}\gamma_\tau(D_i,X_i)+c_{2p}\gamma_\tau(D_j,X_j))\right],$$

so by Lemma 1 and equation (2.16) in Escanciano and Terschuur (2023) we have that

$$\phi^{\gamma}_{ab}(D_i,X_i,D_j,X_j,e,\alpha^{\gamma}) = \sum_{p=1}^{P}\alpha^{\gamma}_{ab,p}(D_i,X_i,D_j,X_j)(c_{1p}Y_i+c_{2p}Y_j-c_{1p}\gamma(D_i,X_i)-c_{2p}\gamma(D_j,X_j)).$$

For the second term notice that

$$\begin{aligned}\mathbb{E}[\varphi(a,X_i,b,X_j,\gamma)] &= \mathbb{E}\left[\varphi(a,X_i,b,X_j,\gamma)\frac{D^{ab}_{ij}}{e_{ab}(X_i,X_j)}\right]\\ &= \mathbb{E}\left[\varphi(D_i,X_i,D_j,X_j,\gamma)\frac{D^{ab}_{ij}}{e_{ab}(X_i,X_j)}\right].\end{aligned}$$

So by the same arguments

$$\phi^{\varphi}_{ab}(D_i,X_i,D_j,X_j,\varphi,\alpha^{\varphi}) = \alpha^{\varphi}_{ab}(D_i,X_i,D_j,X_j)(g(Y_i,X_i,Y_j,X_j,\gamma_a,\gamma_b)-\varphi(D_i,X_i,D_j,X_j,\gamma)),$$

with $\alpha^{\varphi}_{ab}(D_i,X_i,D_j,X_j) = D^{ab}_{ij}/e_{ab}(X_i,X_j)$. ■

Proof of Proposition 3: Let $\gamma_{c,\tau} = \gamma_c + \tau\tilde{\gamma}_c$ for some $\tilde{\gamma}_c \in L_2$. We have that for $(a,b) \in \{0,1\}^2$

$$\begin{aligned}\frac{d}{d\tau}\mathbb{E}[\gamma_{a,\tau}(X_i)+\gamma_{b,\tau}(X_j)] &= \frac{d}{d\tau}\mathbb{E}\left[\gamma_{a,\tau}(X_i)\frac{\mathbb{1}(D_i=a)}{e_a(X_i)}+\gamma_{b,\tau}(X_j)\frac{\mathbb{1}(D_j=b)}{e_b(X_j)}\right]\\ &= \frac{d}{d\tau}\mathbb{E}\left[\bar{\gamma}_\tau(D_i,X_i)\frac{\mathbb{1}(D_i=a)}{e_a(X_i)}+\bar{\gamma}_\tau(D_j,X_j)\frac{\mathbb{1}(D_j=b)}{e_b(X_j)}\right]\\ &= \frac{d}{d\tau}\mathbb{E}_\tau\left[\bar{\gamma}_\tau(D_i,X_i)\frac{\mathbb{1}(D_i=a)}{e_a(X_i)}+\bar{\gamma}_\tau(D_j,X_j)\frac{\mathbb{1}(D_j=b)}{e_b(X_j)}\right]\\ &\quad-\frac{d}{d\tau}\mathbb{E}_\tau\left[\gamma(D_i,X_i)\frac{\mathbb{1}(D_i=a)}{e_a(X_i)}+\gamma(D_j,X_j)\frac{\mathbb{1}(D_j=b)}{e_b(X_j)}\right]\\ &= \frac{d}{d\tau}\mathbb{E}_\tau\left[\frac{\mathbb{1}(D_i=a)}{e_a(X_i)}(Y_i-\gamma(D_i,X_i))+\frac{\mathbb{1}(D_j=b)}{e_b(X_j)}(Y_j-\gamma(D_j,X_j))\right].\end{aligned}$$

Also, let $\Delta_{a,b} = \gamma_a(X_i) - \gamma_b(X_j)$, then

$$\frac{d}{d\tau}\mathbb{E}[|\gamma_{a,\tau}(X_i) - \gamma_{b,\tau}(X_j)|] = \frac{d}{d\tau}\mathbb{E}[|\Delta_{ab} + \tau(\tilde{\gamma}_a(X_i) - \tilde{\gamma}_b(X_j))|].$$

As shown in Escanciano and Terschuur (2023), the Gateaux derivative of the mapping $\Delta \mapsto \mathbb{E}(|\Delta|)$ is some direction ν (assuming no point mass at zero, which follows from the assumptions in the Proposition) is $\mathbb{E}[sgn(\Delta)\nu]$. Hence, by the chain rule

$$\begin{aligned}
&\frac{d}{d\tau}\mathbb{E}[\gamma_{a,\tau}(X_i) + \gamma_{b,\tau}(X_j)] = \frac{d}{d\tau}\mathbb{E}[sgn(\gamma_a(X_i) - \gamma_b(X_j))(\gamma_{a,\tau}(X_i) - \gamma_{b,\tau}(X_j))] \\
&\quad = \frac{d}{d\tau}\mathbb{E}\left[sgn(\gamma_a(X_i) - \gamma_b(X_j))\left(\gamma_{a,\tau}(X_i)\frac{\mathbb{1}(D_i = a)}{e_a(X_i)} - \gamma_{b,\tau}(X_j)\frac{\mathbb{1}(D_j = b)}{e_b(X_j)}\right)\right] \\
&\quad = \frac{d}{d\tau}\mathbb{E}\left[sgn(\gamma_a(X_i) - \gamma_b(X_j))\left(\gamma_\tau(D_i, X_i)\frac{\mathbb{1}(D_i = a)}{e_a(X_i)} - \gamma_\tau(D_j, X_j)\frac{\mathbb{1}(D_j = b)}{e_b(X_j)}\right)\right] \\
&\quad = \frac{d}{d\tau}\mathbb{E}_\tau\left[sgn(\gamma_a(X_i) - \gamma_b(X_j))\left(\gamma_\tau(D_i, X_i)\frac{\mathbb{1}(D_i = a)}{e_a(X_i)} - \gamma_\tau(D_j, X_j)\frac{\mathbb{1}(D_j = b)}{e_b(X_j)}\right)\right] \\
&\quad - \frac{d}{d\tau}\mathbb{E}_\tau\left[sgn(\gamma_a(X_i) - \gamma_b(X_j))\left(\gamma(D_i, X_i)\frac{\mathbb{1}(D_i = a)}{e_a(X_i)} - \gamma(D_j, X_j)\frac{\mathbb{1}(D_j = b)}{e_b(X_j)}\right)\right] \\
&\quad = \frac{d}{d\tau}\mathbb{E}_\tau\left[sgn(\gamma_a(X_i) - \gamma_b(X_j))\left(\frac{\mathbb{1}(D_i = a)}{e_a(X_i)}(Y_i - \gamma(D_i, X_i)) - \frac{\mathbb{1}(D_j = b)}{e_b(X_j)}(Y_j - \gamma(D_j, X_j))\right)\right].
\end{aligned}$$

So by the results in Escanciano and Terschuur (2023), the locally robust score is given by

$$\begin{aligned}
2\Gamma_{ij}^{ab} &= \gamma_a(X_i) + \gamma_b(X_j) - |\gamma_a(X_i) - \gamma_b(X_j)| \\
&+ \frac{\mathbb{1}(D_i = a)}{e_a(X_i)}(Y_i - \gamma(D_i, X_i)) + \frac{\mathbb{1}(D_j = b)}{e_b(X_j)}(Y_j - \gamma(D_j, X_j)) \\
&- sgn(\gamma_a(X_i) - \gamma_b(X_j))\left(\frac{\mathbb{1}(D_i = a)}{e_a(X_i)}(Y_i - \gamma(D_i, X_i)) - \frac{\mathbb{1}(D_j = b)}{e_b(X_j)}(Y_j - \gamma(D_j, X_j))\right) \\
&= \gamma_a(X_i) + \gamma_b(X_j) - |\gamma_a(X_i) - \gamma_b(X_j)| \\
&+ (1 - sgn(\gamma_a(X_i) - \gamma_b(X_j)))\frac{\mathbb{1}(D_i = a)}{e_a(X_i)}(Y_i - \gamma(D_i, X_i)) \\
&+ (1 + sgn(\gamma_a(X_i) - \gamma_b(X_j)))\frac{\mathbb{1}(D_j = b)}{e_b(X_j)}(Y_j - \gamma(D_j, X_j)).
\end{aligned}$$

■

Before proving the rest of the main results I introduce a representation of U-statistics which will be very useful for the coming proofs. For any function $f : \mathcal{Z}^2 \to \mathbb{R}$ let $\mathbb{U}_n f(X_i, X_j) = \binom{n}{2}^{-1} \sum_{i<j} f(X_i, X_j)$. Let κ be the permutations of $\{1, ..., n\}$, then, as in Hoeffding (1963), we can rewrite

$$\mathbb{U}_n f(Z_i, Z_j) = \frac{1}{n!} \sum_{\kappa} \lfloor n/2 \rfloor^{-1} \sum_{i=1}^{\lfloor n/2 \rfloor} f(Z_{\kappa(i)}, Z_{\kappa(\lfloor n/2 \rfloor + i)}). \tag{8.1}$$

This expresses $\mathbb{U}_n f(Z_i, Z_j)$ as a (dependent) sum of averages of i.i.d. random variables (i.e. $f(Z_{\kappa(i)}, Z_{\kappa(\lfloor n/2 \rfloor + i)})$ are i.i.d. for $i = 1, ..., \lfloor n/2 \rfloor$).

Proof of Lemma 2: Using the definition of $W(\pi)$ and $\widetilde{W}_n(\pi)$ and the triangle inequality we know that

$$\begin{aligned}
\mathbb{E}\left[\sup_{\pi \in \Pi} |\widetilde{W}_n(\pi) - W(\pi)|\right] &= \mathbb{E}\left[\sup_{\pi \in \Pi} \left| \mathbb{U}_n \sum_{(a,b) \in \{0,1\}^2} \left(\Gamma_{ij}^{ab} \pi_{ab}(X_i, X_j) - \mathbb{E}[\Gamma_{ij}^{ab} \pi_{ab}(X_i, X_j)] \right) \right| \right] \\
&\leq \sum_{(a,b) \in \{0,1\}^2} \mathbb{E}\left[\sup_{\pi \in \Pi} \left| \mathbb{U}_n \left(\Gamma_{ij}^{ab} \pi_{ab}(X_i, X_j) - \mathbb{E}[\Gamma_{ij}^{ab} \pi_{ab}(X_i, X_j)] \right) \right| \right].
\end{aligned}$$

By the representation used in (8.1) we can rewrite the above as

$$\begin{aligned}
\sum_{(a,b) \in \{0,1\}^2} \mathbb{E}\Bigg[\sup_{\pi \in \Pi} \Bigg| \frac{1}{n!} \sum_{\kappa} \lfloor n/2 \rfloor^{-1} \sum_{i=1}^{\lfloor n/2 \rfloor} \Big(& \Gamma_{\kappa(i)\kappa(\lfloor n/2 \rfloor + i)}^{ab} \pi_{ab}(X_{\kappa(i)}, X_{\kappa(\lfloor n/2 \rfloor + i)}) \\
& - \mathbb{E}[\Gamma_{\kappa(i)\kappa(\lfloor n/2 \rfloor + i)}^{ab} \pi_{ab}(X_{\kappa(i)}, X_{\kappa(\lfloor n/2 \rfloor + i)})] \Big) \Bigg| \Bigg].
\end{aligned} \tag{8.2}$$

Introduce an independent ghost sample $(Z'_1, ..., Z'_n)$ which is distributed as $(Z_1, ..., Z_n)$, Rademacher random variables ε_i, $i = 1, ..., n$, such that $\mathbb{P}(\varepsilon_i = 1) = \mathbb{P}(\varepsilon_i = -1) = 1/2$ and construct ghost scores $\Gamma_{ij}^{\prime ab}$ using the ghost sample. Let $\mathbb{E}_Z$ be the expectation with respect to the distribution of the sample $(Z_1, ..., Z_n)$ and define $\mathbb{E}_{Z'}$ and $\mathbb{E}_\varepsilon$ similarly. Define the Rademacher complexity as

$$\mathcal{R}_n(\Pi) = \mathbb{E}_\varepsilon \left(\sup_{\pi \in \Pi} \left| \lfloor n/2 \rfloor^{-1} \sum_{i=1}^{\lfloor n/2 \rfloor} \varepsilon_i \Gamma_{i, \lfloor n/2 \rfloor + i}^{ab} \pi_{ab}(X_i, X_{\lfloor n/2 \rfloor + i}) \right| \right).$$

Again the key here is that the summands of the sum inside the expectation in $\mathcal{R}_n(\Pi)$ are independent. We are now ready to use a classical symmetrization argument, since

Z_i' has the same distribution as Z_i we have that (8.2) is equal to

$$\sum_{(a,b)\in\{0,1\}^2} \mathbb{E}_Z\left[\sup_{\pi\in\Pi}\left|\frac{1}{n!}\sum_{\kappa}\lfloor n/2\rfloor^{-1}\sum_{i=1}^{\lfloor n/2\rfloor}\left(\Gamma^{ab}_{\kappa(i)\kappa(\lfloor n/2\rfloor+i)}\pi_{ab}(X_{\kappa(i)},X_{\kappa(\lfloor n/2\rfloor+i)})\right.\right.\right.$$

$$\left.\left.\left.-\mathbb{E}_{Z'}[\Gamma'^{ab}_{\kappa(i)\kappa(\lfloor n/2\rfloor+i)}\pi_{ab}(X'_{\kappa(i)},X'_{\kappa(\lfloor n/2\rfloor+i)})]\right)\right|\right]$$

$$\leq \frac{1}{n!}\sum_{\kappa}\sum_{(a,b)\in\{0,1\}^2}\mathbb{E}_{Z,Z'}\left[\sup_{\pi\in\Pi}\left|\lfloor n/2\rfloor^{-1}\sum_{i=1}^{\lfloor n/2\rfloor}\left(\Gamma^{ab}_{\kappa(i)\kappa(\lfloor n/2\rfloor+i)}\pi_{ab}(X_{\kappa(i)},X_{\kappa(\lfloor n/2\rfloor+i)})\right.\right.\right.$$

$$\left.\left.\left.-\Gamma'^{ab}_{\kappa(i)\kappa(\lfloor n/2\rfloor+i)}\pi_{ab}(X'_{\kappa(i)},X'_{\kappa(\lfloor n/2\rfloor+i)})\right)\right|\right]$$

$$=\sum_{(a,b)\in\{0,1\}^2}\mathbb{E}_{Z,Z',\varepsilon}\left[\sup_{\pi\in\Pi}\left|\lfloor n/2\rfloor^{-1}\sum_{i=1}^{\lfloor n/2\rfloor}\varepsilon_i\left(\Gamma^{ab}_{i,\lfloor n/2\rfloor+i}\pi_{ab}(X_i,X_{\lfloor n/2\rfloor+i})\right.\right.\right.$$

$$\left.\left.\left.-\Gamma'^{ab}_{i,\lfloor n/2\rfloor+i}\pi_{ab}(X'_i,X'_{\lfloor n/2\rfloor+i})\right)\right|\right]$$

$$\leq\sum_{(a,b)\in\{0,1\}^2}\mathbb{E}_{Z,Z',\varepsilon}\left[\sup_{\pi\in\Pi}\left|\lfloor n/2\rfloor^{-1}\sum_{i=1}^{\lfloor n/2\rfloor}\varepsilon_i\Gamma^{ab}_{i,\lfloor n/2\rfloor+i}\pi_{ab}(X_i,X_{\lfloor n/2\rfloor+i})\right|\right.$$

$$\left.+\left|\lfloor n/2\rfloor^{-1}\sum_{i=1}^{\lfloor n/2\rfloor}\varepsilon_i\Gamma'^{ab}_{i,\lfloor n/2\rfloor+i}\pi_{ab}(X'_i,X'_{\lfloor n/2\rfloor+i})\right|\right]$$

$$=\sum_{(a,b)\in\{0,1\}^2}\mathbb{E}[2\mathcal{R}_n(\Pi)].$$

The first inequality follows from Jensen's and triangle inequalities, the second equality uses the fact that the vector $(Z_{\pi(i)}, Z_{\pi(\lfloor n/2\rfloor+i)}, Z'_{\pi(i)}, Z'_{\pi(\lfloor n/2\rfloor+i)})$ is identically distributed across $i=1,...,\lfloor n/2\rfloor$ for all permutations in κ (so we can take the permutation $\kappa(i)=i$) and the fact that $\varepsilon_i(\Gamma^{ab}_{i,\lfloor n/2\rfloor+i}\pi_{ab}(X_i,X_{\lfloor n/2\rfloor+i})-\Gamma'^{ab}_{i,\lfloor n/2\rfloor+i}\pi_{ab}(X'_i,X'_{\lfloor n/2\rfloor+i}))$ and $\Gamma^{ab}_{i,\lfloor n/2\rfloor+i}\pi_{ab}(X_i,X_{\lfloor n/2\rfloor+i})-\Gamma'^{ab}_{i,\lfloor n/2\rfloor+i}\pi_{ab}(X'_i,X'_{\lfloor n/2\rfloor+i})$ have the same distribution, the third inequality uses the triangle inequality and the last equality uses that $Z_i\sim Z'_i$ and the definition of the Rademacher complexity. ■

Proof of Lemma 3: Note that Lemma 6 gives us a sequence of covers B_k for $k=0,...,K$ of $\tilde{\Pi}_{ab}$ of radius less than 2^{-k} for some K. For any $j=1,...,J$ with $J=\lceil\log_2(\lfloor n/2\rfloor)(1-\beta)\rceil$ and $\pi\in\tilde{\Pi}_{ab}$ let $b_j:\tilde{\Pi}_{ab}\mapsto\tilde{\Pi}_{ab}$ be an operator such that $b_j(\pi)$ is an approximating policy from the cover B_j such that $D_n(\pi,b_j(\pi))\leq 2^{-j}$, such an approximation exists by Lemma 6. By the same Lemma we also know that $|\{b_j(\pi):$

$\pi \in \tilde{\Pi}_{ab}\}| \leq N_{D_n}(2^{-j}, \widetilde{\Pi}_{ab}, \{X_i, \Gamma_{i,\lfloor n/2\rfloor+i}\}_{i=1}^{\lfloor n/2\rfloor})$. Let $\underline{J} = \lceil 1/2 \log_2(\lfloor n/2\rfloor)(1-\beta)\rceil$. By using a telescope sum and the approximations $b_0, ..., b_J$ we can decompose the Rademacher complexity as

$$\begin{aligned}\mathcal{R}_n(\Pi) = \mathbb{E}_\varepsilon \Bigg\{ \sup_{\pi \in \Pi} \Bigg| \lfloor n/2\rfloor^{-1} \sum_{i=1}^{\lfloor n/2\rfloor} \varepsilon_i \Gamma_{i,\lfloor n/2\rfloor+i}^{ab} \Bigg[& b_0(\pi_{ab}(X_i, X_{\lfloor n/2\rfloor+i})) \\ & + \sum_{j=1}^{\underline{J}} \Big(b_j(\pi_{ab}(X_i, X_{\lfloor n/2\rfloor+i})) - b_{j-1}(\pi_{ab}(X_i, X_{\lfloor n/2\rfloor+i})) \Big) \\ & + (b_J(\pi_{ab}(X_i, X_{\lfloor n/2\rfloor+i})) - b_{\underline{J}}(\pi_{ab}(X_i, X_{\lfloor n/2\rfloor+i}))) \\ & + (\pi_{ab}(X_i, X_{\lfloor n/2\rfloor+i}) - b_J(\pi_{ab}(X_i, X_{\lfloor n/2\rfloor+i}))) \Bigg] \Bigg| \Bigg\}.\end{aligned}$$

Note that since the distance D_n is bounded by 1, by the second property in Lemma 6 we have that b_0 can be any policy in $\tilde{\Pi}_{ab}$. Hence, we can set $b_0(\pi_{ab}(X_i, X_{\lfloor n/2\rfloor+i})) = 0$ for all $i = 1, ..., \lfloor n/2\rfloor$. We approach each of the terms above in turn. Note that $b_0, ..., b_J$ is a sequence of increasingly accurate approximations. The first step is to notice that the last term above is negligible, i.e. the term involving the closest approximation vanishes at a $\sqrt{n}$ rate. By using Cauchy-Schwarz and multiplying and dividing we get

$$\begin{aligned}&\sqrt{\lfloor n/2\rfloor} \sup_{\pi \in \Pi} \Bigg| \lfloor n/2\rfloor^{-1} \sum_{i=1}^{\lfloor n/2\rfloor} \varepsilon_i \Gamma_{i,\lfloor n/2\rfloor+i}^{ab} (\pi_{ab}(X_i, X_{\lfloor n/2\rfloor+i}) - b_J(\pi_{ab}(X_i, X_{\lfloor n/2\rfloor+i}))) \Bigg| \\ &\leq \sqrt{\lfloor n/2\rfloor} \sup_{\pi \in \Pi} \frac{\sqrt{\lfloor n/2\rfloor^{-1} \sum_{i=1}^{\lfloor n/2\rfloor} \Big| \Gamma_{i,\lfloor n/2\rfloor+i}^{ab} (\pi_{ab}(X_i, X_{\lfloor n/2\rfloor+i}) - b_J(\pi_{ab}(X_i, X_{\lfloor n/2\rfloor+i}))) \Big|^2}}{\sqrt{\lfloor n/2\rfloor^{-1} \sum_{i=1}^{\lfloor n/2\rfloor} \Gamma_{i,\lfloor n/2\rfloor+i}^{2\,ab}}} \\ &\times \sqrt{\lfloor n/2\rfloor^{-1} \sum_{i=1}^{\lfloor n/2\rfloor} \Gamma_{i,\lfloor n/2\rfloor+i}^{2\,ab}} \\ &= \sqrt{\lfloor n/2\rfloor} \sup_{\pi \in \Pi} D_n(\pi_{ab}, b_J(\pi_{ab})) \sqrt{\lfloor n/2\rfloor^{-1} \sum_{i=1}^{\lfloor n/2\rfloor} \Gamma_{i,\lfloor n/2\rfloor+i}^{2\,ab}} \\ &\leq \sqrt{\lfloor n/2\rfloor} 2^{-J} \sqrt{\lfloor n/2\rfloor^{-1} \sum_{i=1}^{\lfloor n/2\rfloor} \Gamma_{i,\lfloor n/2\rfloor+i}^{2\,ab}} = \frac{M}{\lfloor n/2\rfloor^{1/2-\beta}} \to 0,\end{aligned}$$

last inequality uses Lemma 6 and the last equality uses that $J = \lceil \log_2(\lfloor n/2 \rfloor)(1-\beta) \rceil$ and the boundedness assumption. Now we show that the second to last term of the Rademacher decomposition is also negligible. Notice that $\{\varepsilon_i \Gamma^{ab}_{i,\lfloor n/2\rfloor+i}(b_J(\pi_{ab}(X_i, X_{\lfloor n/2\rfloor+i})) - b_{\underline{J}}(\pi_{ab}(X_i, X_{\lfloor n/2\rfloor+i})))\}_{i=1}^{\lfloor n/2\rfloor}$ are zero mean (conditional on $\{X_i, \Gamma_{i,\lfloor n/2\rfloor+i}\}_{i=1}^{\lfloor n/2\rfloor}$) i.i.d. random variables. They are also bounded below by $a_i = -|\Gamma^{ab}_{i,\lfloor n/2\rfloor+i}(b_J(\pi_{ab}(X_i, X_{\lfloor n/2\rfloor+i})) - b_{\underline{J}}(\pi_{ab}(X_i, X_{\lfloor n/2\rfloor+i})))|$ and above by $b_i = -a_i$. Hence, by Hoeffding's inequality

$$\begin{aligned}
&\mathbb{P}_\varepsilon\left(\left|\sum_{i=1}^{\lfloor n/2\rfloor} \varepsilon_i \Gamma^{ab}_{i,\lfloor n/2\rfloor+i}(b_J(\pi_{ab}(X_i, X_{\lfloor n/2\rfloor+i})) - b_{\underline{J}}(\pi_{ab}(X_i, X_{\lfloor n/2\rfloor+i})))\right| \geq t\right) \\
&\quad \leq 2\exp\left(-\frac{2t^2}{\sum_{i=1}^{\lfloor n/2\rfloor}(b_i - a_i)^2}\right) = 2\exp\left(-\frac{t^2}{D_n^2(b_J(\pi_{ab}), b_{\underline{J}}(\pi_{ab}))\sum_{i=1}^{\lfloor n/2\rfloor}\Gamma^{2\,ab}_{i,\lfloor n/2\rfloor+i}}\right).
\end{aligned}$$

Hence, for any $a > 0$ we have that

$$\begin{aligned}
&\mathbb{P}_\varepsilon\Bigg(\left|\sqrt{\lfloor n/2\rfloor}\lfloor n/2\rfloor^{-1}\sum_{i=1}^{\lfloor n/2\rfloor} \varepsilon_i \Gamma^{ab}_{i,\lfloor n/2\rfloor+i}(b_J(\pi_{ab}(X_i, X_{\lfloor n/2\rfloor+i})) - b_{\underline{J}}(\pi_{ab}(X_i, X_{\lfloor n/2\rfloor+i})))\right| \\
&\quad \geq a2^{2-\underline{J}}\sqrt{\frac{\sum_{i=1}^{\lfloor n/2\rfloor}\Gamma^{2\,ab}_{i,\lfloor n/2\rfloor+i}}{\lfloor n/2\rfloor}}\Bigg) \\
&\quad \leq 2\exp\left(-\frac{a^2 4^{2-\underline{J}}}{D_n^2(b_J(\pi_{ab}), b_{\underline{J}}(\pi_{ab}))}\right) \\
&\quad \leq 2\exp\left(-\frac{a^2 4^{2-\underline{J}}}{\sum_{j=\underline{J}}^{J-1} D_n^2(b_j(\pi_{ab}), b_{j+1}(\pi_{ab}))}\right) \\
&\quad \leq 2\exp\left(-\frac{a^2 4^{2-\underline{J}}}{\left(\sum_{j=\underline{J}}^{J-1} 2^{-(j-1)}\right)^2}\right) \\
&\quad \leq 2\exp(-a^2),
\end{aligned}$$

where we have used triangle inequality in the second inequality and the fact that $\sum_{j=\underline{J}}^{J-1} 2^{-(j-1)} = 2^{2-\underline{J}} - 2^{2-J} \leq 2^{2-\underline{J}}$ in the last inequality. This holds for any policy, hence

$$\begin{aligned}
&\mathbb{P}_\varepsilon\Bigg(\sup_{\pi\in\Pi}\Bigg|\lfloor n/2\rfloor^{-1/2}\sum_{i=1}^{\lfloor n/2\rfloor}\varepsilon_i\Gamma^{ab}_{i,\lfloor n/2\rfloor+i}(b_J(\pi_{ab}(X_i,X_{\lfloor n/2\rfloor+i}))-b_{\underline{J}}(\pi_{ab}(X_i,X_{\lfloor n/2\rfloor+i})))\Bigg|\\
&\quad\geq a2^{2-\underline{J}}\sqrt{\frac{\sum_{i=1}^{\lfloor n/2\rfloor}\Gamma^{2\,ab}_{i,\lfloor n/2\rfloor+i}}{\lfloor n/2\rfloor}}\Bigg)\\
&\quad\leq 2|\{b_J(\pi_{ab}),b_{\underline{J}}(\pi_{ab})\}|\exp(-a^2)\\
&\quad\leq 2N_{D_n}(2^{-J},\widetilde{\Pi}_{ab},\{X_i,\Gamma_{i,\lfloor n/2\rfloor+i}\}_{i=1}^{\lfloor n/2\rfloor})\exp(-a^2)\\
&\quad\leq 2N_H(2^{-2J},\widetilde{\Pi}_{ab})\exp(-a^2)\\
&\quad= 2\exp(\log(N_H(2^{-2J},\widetilde{\Pi}_{ab})))\exp(-a^2)\\
&\quad\leq 2\exp(5VC(\widetilde{\Pi}_{ab})\log(2^{2J})-a^2)\\
&\quad\leq 2\exp(5VC(\widetilde{\Pi}_{ab})\log(2^{-2(1-\beta)\log_2(\lfloor n/2\rfloor)})-a^2),
\end{aligned}$$

where in the first inequality I use the union bound, in the second inequality I use properties of the approximations (see Zhou et al. (2023)), in the third I use Lemma 5 and in the fourth inequality I bound the log of the Hamming covering number by the VC dimension using a result in Haussler (1995). Let now

$$a=\frac{2^{\underline{J}}}{\sqrt{\log(\lfloor n/2\rfloor)\lfloor n/2\rfloor^{-1}\sum_{i=1}^{\lfloor n/2\rfloor}\Gamma^{2\,ab}_{i,\lfloor n/2\rfloor+i}}}\text{ so that }a2^{2-\underline{J}}\sqrt{\frac{\sum_{i=1}^{\lfloor n/2\rfloor}\Gamma^{2\,ab}_{i,\lfloor n/2\rfloor+i}}{\lfloor n/2\rfloor}}=\frac{4}{\sqrt{\log(\lfloor n/2\rfloor)}}.$$

Finally,

$$\begin{aligned}
&\mathbb{P}_\varepsilon\Bigg(\sup_{\pi\in\Pi}\Bigg|\lfloor n/2\rfloor^{-1/2}\sum_{i=1}^{\lfloor n/2\rfloor}\varepsilon_i\Gamma^{ab}_{i,\lfloor n/2\rfloor+i}(b_J(\pi_{ab}(X_i,X_{\lfloor n/2\rfloor+i}))-b_{\underline{J}}(\pi_{ab}(X_i,X_{\lfloor n/2\rfloor+i})))\Bigg|\\
&\geq\frac{4}{\sqrt{\log(\lfloor n/2\rfloor)}}\Bigg)\leq 2\exp\Bigg(5VC(\widetilde{\Pi}_{ab})\log(\lfloor n/2\rfloor^{-2(1-\beta)})-\frac{\lfloor n/2\rfloor^{-\beta}}{\log(\lfloor n/2\rfloor)\sum_{i=1}^{\lfloor n/2\rfloor}\Gamma^{2\,ab}_{i,\lfloor n/2\rfloor+i}}\Bigg)\\
&\leq 2\exp\Bigg\{-5\lfloor n/2\rfloor^{\beta}\log\Big(\lfloor n/2\rfloor^{2(1-\beta)}\Big)-\frac{1}{\lfloor n/2\rfloor^{\beta}\log(\lfloor n/2\rfloor)M^2}\Bigg\}\to 0,
\end{aligned}$$

where I have used Assumption 7 and the boundedness assumption.

$$\mathbb{E}\Bigg(\sup_{\pi\in\Pi}\Bigg|\lfloor n/2\rfloor^{-1/2}\sum_{i=1}^{\lfloor n/2\rfloor}\varepsilon_i\Gamma^{ab}_{i,\lfloor n/2\rfloor+i}(b_J(\pi_{ab}(X_i,X_{\lfloor n/2\rfloor+i}))-b_{\underline{J}}(\pi_{ab}(X_i,X_{\lfloor n/2\rfloor+i})))\Bigg|\Bigg)\to 0,$$

since for any sequence of random variables X_n and sequence of real numbers a_n if $\lim_{n\to\infty} \mathbb{P}(X_n \le a_n) = 1$ and $\lim_{n\to\infty} a_n = 0$, then $\lim_{n\to\infty} \mathbb{E}(X_n) = 0$ (proof of this fact uses $\mathbb{E}(X_n) = \int_0^\infty \mathbb{P}(X_n > u)\, du$). Hence, we have proven that

$$\mathbb{E}[\mathcal{R}_n(\Pi)] = \mathbb{E}\left\{\sup_{\pi\in\Pi}\left|\lfloor n/2\rfloor^{-1}\sum_{i=1}^{\lfloor n/2\rfloor}\varepsilon_i\Gamma^{ab}_{i,\lfloor n/2\rfloor+i}\left[\sum_{j=1}^{\underline{J}}\Big(b_j(\pi_{ab}(X_i, X_{\lfloor n/2\rfloor+i})) - b_{j-1}(\pi_{ab}(X_i, X_{\lfloor n/2\rfloor+i}))\Big)\right]\right|\right\} + o\left(\frac{1}{\sqrt{n}}\right).$$

Hence I have left what Zhou et al. (2023) call the effective regime. Let $j \in \{1, ..., \underline{J}\}$ and a_j be some constant depending on j. As before, conditional on $\{X_i, \Gamma_{i,\lfloor n/2\rfloor+i}\}_{i=1}^{\lfloor n/2\rfloor}$ we can apply Hoeffding inequality and then use the definition of D_n to get

$$\begin{aligned}
&\mathbb{P}_\varepsilon\Bigg(\left|\lfloor n/2\rfloor^{-1/2}\sum_{i=1}^{\lfloor n/2\rfloor}\varepsilon_i\Gamma^{ab}_{i,\lfloor n/2\rfloor+i}(b_j(\pi_{ab}(X_i, X_{\lfloor n/2\rfloor+i})) - b_{j-1}(\pi_{ab}(X_i, X_{\lfloor n/2\rfloor+i})))\right| \\
&\quad \ge a_j 2^{2-j}\sqrt{\frac{\sum_{i=1}^{\lfloor n/2\rfloor}\Gamma^{2\,ab}_{i,\lfloor n/2\rfloor+i}}{\lfloor n/2\rfloor}}\Bigg) \\
&\quad \le 2\exp\left(-\frac{a_j^2 4^{2-j}}{D_n^2(b_j(\pi_{ab}), b_{j-1}(\pi_{ab}))}\right) \\
&\quad \le 2\exp\left(\frac{-a_j^2 4^{2-j}}{4^{-(j-1)}}\right) \\
&\quad = 2\exp\left(-4a_j^2\right),
\end{aligned}$$

the last inequality uses the fact that $D_n(b_j(\pi_{ab}), b_{j-1}(\pi_{ab})) \le 2^{-(j-1)}$ by Lemma 6. Now we let

$$a_j^2(k) = 2\log\left(\frac{2j^2}{\delta_k}N_H(4^{-j}, \widetilde{\Pi}_{ab})\right),$$

where δ_k is some sequence of real numbers indexed by $k \in \mathbb{N}$. For notational convenience define

$$R_j = \sup_{\pi\in\Pi}\left|\lfloor n/2\rfloor^{-1/2}\sum_{i=1}^{\lfloor n/2\rfloor}\varepsilon_i\Gamma^{ab}_{i,\lfloor n/2\rfloor+i}(b_j(\pi_{ab}(X_i, X_{\lfloor n/2\rfloor+i})) - b_{j-1}(\pi_{ab}(X_i, X_{\lfloor n/2\rfloor+i})))\right|.$$

Then we have that

$$\begin{aligned}
\mathbb{P}\Bigg(R_j \geq a_j(k)2^{-j}\sqrt{\lfloor n/2\rfloor^{-1}\sum_{i=1}^{\lfloor n/2\rfloor}\Gamma^{2\,ab}_{i,\lfloor n/2\rfloor+i}}\Bigg) &\leq 2|\{b_j(\pi_{ab}), b_{j-1}(\pi_{ab})\}|\exp(-a_j^2(k)/2) \\
&\leq 2N_{D_n}(2^{-j}, \widetilde{\Pi}_{ab}, \{X_i, \Gamma_{i,\lfloor n/2\rfloor+i}\}_{i=1}^{\lfloor n/2\rfloor})\exp(-a_j^2(k)/2) \\
&\leq 2N_H(2^{-2j}, \widetilde{\Pi}_{ab})\exp(-a_j^2(k)/2) \\
&= 2N_H(4^{-j}, \widetilde{\Pi}_{ab})\exp(-\log(N_H(4^{-j}, \widetilde{\Pi}_{ab})2j^2/\delta_k)) \\
&= \frac{\delta_k}{j^2}.
\end{aligned}$$

Sum across j and apply this bound with $\delta_k = 1/2^k$ to note that

$$\begin{aligned}
\sum_{j=1}^{J}\mathbb{P}\Bigg(R_j \geq a_j(k)2^{-j}\sqrt{\lfloor n/2\rfloor^{-1}\sum_{i=1}^{\lfloor n/2\rfloor}\Gamma^{2\,ab}_{i,\lfloor n/2\rfloor+i}}\Bigg) &\leq \sum_{j=1}^{J}\frac{\delta_k}{j^2} \\
&\leq \sum_{j=1}^{\infty}\frac{\delta_k}{j^2} \\
&\leq \frac{1.7}{2^k}.
\end{aligned}$$

Let F_{R_j} be the cumulative distribution function of R_j (conditional on $\{X_i, \Gamma_{i,\lfloor n/2\rfloor+i}\}_{i=1}^{\lfloor n/2\rfloor}$).

We can bound the following object of interest in the following way

$$
\begin{aligned}
&\mathbb{E}_\varepsilon\left[\sup_{\pi\in\Pi}\left|\lfloor n/2\rfloor^{-1/2}\sum_{i=1}^{\lfloor n/2\rfloor}\varepsilon_i\Gamma^{ab}_{i,\lfloor n/2\rfloor+i}\sum_{j=1}^{J}(b_j(\pi_{ab}(X_i,X_{\lfloor n/2\rfloor+i}))-b_{j-1}(\pi_{ab}(X_i,X_{\lfloor n/2\rfloor+i})))\right|\right] \\
&\quad\le\sum_{j=1}^{J}\mathbb{E}_\varepsilon[R_j]=\int_0^\infty\sum_{j=1}^{J}(1-F_{R_j}(r))\,dr\le\int_0^\infty\sum_{j=1}^{J}\mathbb{P}(R_j\ge r)\,dr \\
&\quad\le\sum_{k=0}^{\infty}\sum_{j=1}^{J}\frac{1.7}{2^k}a_j(k)2^{-j}\sqrt{\lfloor n/2\rfloor^{-1}\sum_{i=1}^{\lfloor n/2\rfloor}\Gamma^{2\,ab}_{i,\lfloor n/2\rfloor+i}} \\
&\quad\le\sum_{k=0}^{\infty}\sum_{j=1}^{J}\frac{1.7}{2^k}\sqrt{2}\sqrt{\log(2^{k+1}j^2N_H(4^{-j},\widetilde{\Pi}_{ab}))}2^{-j}\sqrt{\lfloor n/2\rfloor^{-1}\sum_{i=1}^{\lfloor n/2\rfloor}\Gamma^{2\,ab}_{i,\lfloor n/2\rfloor+i}} \\
&\quad\le 1.7\sqrt{2}\sqrt{\lfloor n/2\rfloor^{-1}\sum_{i=1}^{\lfloor n/2\rfloor}\Gamma^{2\,ab}_{i,\lfloor n/2\rfloor+i}}\sum_{k=0}^{\infty}2^{-k}\sum_{j=1}^{J}2^{-j}\sqrt{(k+1)\log 2+2\log j+\log N_H(4^{-j},\widetilde{\Pi}_{ab})} \\
&\quad\le 1.7\sqrt{2}\sqrt{\lfloor n/2\rfloor^{-1}\sum_{i=1}^{\lfloor n/2\rfloor}\Gamma^{2\,ab}_{i,\lfloor n/2\rfloor+i}}\sum_{k=0}^{\infty}2^{-k}\sum_{j=1}^{J}2^{-j}\left(\sqrt{k+1}+\sqrt{2\log j}+\sqrt{5VC(\widetilde{\Pi}_{ab})\log(4^j)}\right) \\
&\quad\le 1.7\sqrt{2}\sqrt{\lfloor n/2\rfloor^{-1}\sum_{i=1}^{\lfloor n/2\rfloor}\Gamma^{2\,ab}_{i,\lfloor n/2\rfloor+i}}\sum_{k=0}^{\infty}2^{-k}\Bigg(\sqrt{k+1}\sum_{j=1}^{\infty}2^{-j}+\sqrt{2}\sum_{j=1}^{\infty}2^{-j}\sqrt{\log j} \\
&\quad+\sqrt{5VC(\widetilde{\Pi}_{ab})}\sum_{j=1}^{\infty}2^{-j}\sqrt{\log 4^j}\Bigg) \\
&\quad\le 1.7\sqrt{2}\sqrt{\lfloor n/2\rfloor^{-1}\sum_{i=1}^{\lfloor n/2\rfloor}\Gamma^{2\,ab}_{i,\lfloor n/2\rfloor+i}}\left(\sum_{k=0}^{\infty}2^{-k}\sqrt{k+1}+\frac{\sqrt{2}}{2}\sum_{k=0}^{\infty}2^{-k}+\sqrt{5VC(\widetilde{\Pi}_{ab})}1.6\sum_{k=0}^{\infty}2^{-k}\right) \\
&\quad\le 1.7\sqrt{2}\sqrt{\lfloor n/2\rfloor^{-1}\sum_{i=1}^{\lfloor n/2\rfloor}\Gamma^{2\,ab}_{i,\lfloor n/2\rfloor+i}}\left(5+3.2\sqrt{5VC(\widetilde{\Pi}_{ab})}\right).
\end{aligned}
$$

So taking expectations over $\{X_i,\Gamma_{i,\lfloor n/2\rfloor+i}\}_{i=1}^{\lfloor n/2\rfloor}$, using this bound and the Jensen's inequality we get

$$\begin{aligned}
&\mathbb{E}\left[\sup_{\pi\in\Pi}\left|\lfloor n/2\rfloor^{-1/2}\sum_{i=1}^{\lfloor n/2\rfloor}\varepsilon_i\Gamma^{ab}_{i,\lfloor n/2\rfloor+i}\sum_{j=1}^{J}(b_j(\pi_{ab}(X_i,X_{\lfloor n/2\rfloor+i}))-b_{j-1}(\pi_{ab}(X_i,X_{\lfloor n/2\rfloor+i})))\right|\right]\\
&\quad\leq 1.7\sqrt{2}\left(5+8\sqrt{5VC(\widetilde{\Pi}_{ab})}\right)\mathbb{E}\left[\sqrt{\lfloor n/2\rfloor^{-1}\sum_{i=1}^{\lfloor n/2\rfloor}\Gamma^{2\,ab}_{i,\lfloor n/2\rfloor+i}}\right]\\
&\quad\leq 1.7\sqrt{2}\left(5+8\sqrt{5VC(\widetilde{\Pi}_{ab})}\right)\sqrt{\mathbb{E}\left[\Gamma^{2\,ab}_{i,\lfloor n/2\rfloor+i}\right]}\\
&\quad= 1.7\sqrt{2}\left(5+8\sqrt{5VC(\widetilde{\Pi}_{ab})}\right)\sqrt{S_{ab}}\leq C\sqrt{VC(\widetilde{\Pi}_{ab})S_{ab}},
\end{aligned}$$

for some constant $C>0$. Dividing both sides by $\sqrt{\lfloor n/2\rfloor}$ we get

$$\begin{aligned}
&\mathbb{E}\left[\sup_{\pi\in\Pi}\left|\lfloor n/2\rfloor^{-1}\sum_{i=1}^{\lfloor n/2\rfloor}\varepsilon_i\Gamma^{ab}_{i,\lfloor n/2\rfloor+i}\sum_{j=1}^{J}(b_j(\pi_{ab}(X_i,X_{\lfloor n/2\rfloor+i}))-b_{j-1}(\pi_{ab}(X_i,X_{\lfloor n/2\rfloor+i})))\right|\right]\\
&\quad\leq C\sqrt{\frac{VC(\widetilde{\Pi}_{ab})S_{ab}}{\lfloor n/2\rfloor}},
\end{aligned}$$

and hence

$$\mathbb{E}[\mathcal{R}_n(\Pi)]\leq C\sqrt{\frac{VC(\widetilde{\Pi}_{ab})S_{ab}}{\lfloor n/2\rfloor}}+o\left(\frac{1}{\sqrt{n}}\right)=\mathcal{O}\left(\sqrt{\frac{VC(\widetilde{\Pi}_{ab})S_{ab}}{\lfloor n/2\rfloor}}\right).$$

■

Proof of Lemma 4: Define the following random variables

$$\begin{aligned}
\hat{R}^{(1)}_{ij,ab,l}&=m_{ab}(Z_i,Z_j,\hat{\gamma}_l,\varphi)-m_{ab}(Z_i,Z_j,\gamma,\varphi), & \hat{R}^{(2)}_{ij,ab,l}&=m_{ab}(Z_i,Z_j,\gamma,\hat{\varphi}_l)-m_{ab}(Z_i,Z_j,\gamma,\varphi)\\
\hat{R}^{(3)}_{ij,ab,l}&=\phi^{\gamma}_{ab}(Z_i,Z_j,\hat{\gamma}_l,\alpha^{\gamma})-\phi^{\gamma}_{ab}(Z_i,Z_j,\gamma,\alpha^{\gamma}), & \hat{R}^{(4)}_{ij,ab,l}&=\phi^{\gamma}_{ab}(Z_i,Z_j,\gamma,\hat{\alpha}^{\gamma}_l)-\phi^{\gamma}_{ab}(Z_i,Z_j,\gamma,\alpha^{\gamma})\\
\hat{R}^{(5)}_{ij,ab,l}&=\phi^{\varphi}_{ab}(Z_i,Z_j,\hat{\varphi}_l,\alpha^{\varphi})-\phi^{\varphi}_{ab}(Z_i,Z_j,\varphi,\alpha^{\varphi}), & \hat{R}^{(6)}_{ij,ab,l}&=\phi^{\varphi}_{ab}(Z_i,Z_j,\varphi,\hat{\alpha}^{\varphi}_l)-\phi^{\varphi}_{ab}(Z_i,Z_j,\varphi,\alpha^{\varphi}).
\end{aligned}$$

Then,

$$\mathbb{E}\left[\sup_{\pi\in\Pi_n}|\hat{W}_n(\pi)-\widetilde{W}_n(\pi)|\right]=\mathbb{E}\left(\sup_{\pi\in\Pi_n}\left|\binom{n}{2}^{-1}\sum_{l=1}^{L}\sum_{(i,j)\in I_l}\sum_{(a,b)\in\pi}\sum_{k=1}^{6}(\hat{R}^{(k)}_{ij,ab,l}+\hat{\xi}_{ij,ab,l}+\hat{\xi}^{\gamma}_{ij,ab,l}+\hat{\xi}^{\varphi}_{ij,ab,l})\pi_{ab}(X_i,X_j)\right|\right).$$

By repeated use of the triangle inequality

$$(\dagger)\quad \mathbb{E}\Bigg[\sup_{\pi\in\Pi_n}|\hat{W}_n(\pi)-\widetilde{W}_n(\pi)|\Bigg] \leq \sum_{l=1}^{L}\sum_{(a,b)\in\pi}\mathbb{E}\Bigg(\sup_{\pi\in\Pi_n}\Bigg|\binom{n}{2}^{-1}\sum_{(i,j)\in I_l}(\hat{R}^{(1)}_{ij,l}+\hat{R}^{(3)}_{ij,l})\pi_{ab}(X_i,X_j)\Bigg|\Bigg)$$
$$+\sum_{l=1}^{L}\sum_{(a,b)\in\pi}\mathbb{E}\Bigg(\sup_{\pi\in\Pi_n}\Bigg|\binom{n}{2}^{-1}\sum_{(i,j)\in I_l}(\hat{R}^{(2)}_{ij,l}+\hat{R}^{(5)}_{ij,l})\pi_{ab}(X_i,X_j)\Bigg|\Bigg)$$
$$+\sum_{l=1}^{L}\sum_{(a,b)\in\pi}\mathbb{E}\Bigg(\sup_{\pi\in\Pi_n}\Bigg|\binom{n}{2}^{-1}\sum_{(i,j)\in I_l}(\hat{R}^{(4)}_{ij,l}+\hat{R}^{(6)}_{ij,l})\pi_{ab}(X_i,X_j)\Bigg|\Bigg)$$
$$+\sum_{l=1}^{L}\sum_{(a,b)\in\pi}\mathbb{E}\Bigg(\sup_{\pi\in\Pi_n}\Bigg|\binom{n}{2}^{-1}\sum_{(i,j)\in I_l}(\hat{\xi}_{ij,l}+\hat{\xi}^{\gamma}_{ij,l}+\hat{\xi}^{\varphi}_{ij,l})\pi_{ab}(X_i,X_j)\Bigg|\Bigg).$$

I will bound each of the terms separately. The same arguments apply for all $l=1,...,L$ and $(a,b)\in\pi$, hence we focus on some fixed (a,b) and l. Let N_l^c be the observations not in I_l. By adding and subtracting $\mathbb{E}[(\hat{R}^{(1)}_{ij,ab,l}+\hat{R}^{(3)}_{ij,ab,l})\pi_{ab}(X_i,X_j)|N_l^c]$ and applying the triangle inequality we get that the summands of the first term are bounded by

$$\mathbb{E}\Bigg(\sup_{\pi\in\Pi_n}\Bigg|\binom{n}{2}^{-1}\sum_{(i,j)\in I_l}(\hat{R}^{(1)}_{ij,l}+\hat{R}^{(3)}_{ij,l})\pi_{ab}(X_i,X_j)-\mathbb{E}[(\hat{R}^{(1)}_{ij,ab,l}+\hat{R}^{(3)}_{ij,ab,l})\pi_{ab}(X_i,X_j)|N_l^c]\Bigg|\Bigg)\quad(\star)$$
$$+\mathbb{E}\Bigg(\sup_{\pi\in\Pi_n}\binom{n}{2}^{-1}\sum_{(i,j)\in I_l}|\mathbb{E}[(\hat{R}^{(1)}_{ij,ab,l}+\hat{R}^{(3)}_{ij,ab,l})\pi_{ab}(X_i,X_j)|\hat{\gamma}_l]|\Bigg).\quad(\star\star)$$

By Assumption 5 we know that

$$|\mathbb{E}[\hat{R}^{(1)}_{ij,ab,l}+\hat{R}^{(3)}_{ij,ab,l}|N_l^c]|=|\mathbb{E}[m_{ab}(Z_i,Z_j,\hat{\gamma}_l,\varphi)+\phi^{\gamma}_{ab}(Z_i,Z_j,\hat{\gamma}_l,\alpha^{\gamma})|\hat{\gamma}_l]|\leq C||\hat{\gamma}_l-\gamma||^2.$$

Applying the conditional Jensen's inequality (on the absolute value) in $(\star\star)$ and noting that the resulting expression is maximized by treating everybody we get that

$$(\star\star)\leq C\mathbb{E}[||\hat{\gamma}_l-\gamma||^2]\underbrace{\binom{n}{2}^{-1}|I_l|}_{\leq 1}=o(n^{-2\lambda_\gamma})=o(1/\sqrt{n}),$$

where the last equality follows since $2\lambda_\gamma \geq 1/2$. For $(\star)$, note that

$$
\begin{aligned}
(\star) \leq & \binom{n}{2}^{-1}|I_l|\mathbb{E}\left(\sup_{\pi\in\Pi_n}\left||I_l|^{-1}\sum_{(i,j)\in I_l}\hat{R}^{(1)}_{ij,l}\pi_{ab}(X_i,X_j)-\mathbb{E}[\hat{R}^{(1)}_{ij,ab,l}\pi_{ab}(X_i,X_j)|N_l^c]\right|\right) \\
&+\binom{n}{2}^{-1}|I_l|\mathbb{E}\left(\sup_{\pi\in\Pi_n}\left||I_l|^{-1}\sum_{(i,j)\in I_l}\hat{R}^{(3)}_{ij,l}\pi_{ab}(X_i,X_j)-\mathbb{E}[\hat{R}^{(3)}_{ij,ab,l}\pi_{ab}(X_i,X_j)|N_l^c])\right|\right) \\
&=\binom{n}{2}^{-1}|I_l|\mathbb{E}\left[\mathbb{E}\left(\sup_{\pi\in\Pi_n}\left||I_l|^{-1}\sum_{(i,j)\in I_l}\hat{R}^{(1)}_{ij,l}\pi_{ab}(X_i,X_j)-\mathbb{E}[\hat{R}^{(1)}_{ij,ab,l}\pi_{ab}(X_i,X_j)|N_l^c]\right|\,\middle|N_l^c\right)\right] \\
&+\binom{n}{2}^{-1}|I_l|\mathbb{E}\left[\mathbb{E}\left(\sup_{\pi\in\Pi_n}\left||I_l|^{-1}\sum_{(i,j)\in I_l}\hat{R}^{(3)}_{ij,l}\pi_{ab}(X_i,X_j)-\mathbb{E}[\hat{R}^{(3)}_{ij,ab,l}\pi_{ab}(X_i,X_j)|N_l^c]\right|\,\middle|N_l^c\right)\right].
\end{aligned}
$$

The inner expectations are the expected supremum of centered U-processes. Using Lemma 2 We can bound these inner expectations with Rademacher complexities. However, in the same way we used the construction in Equation (8.1) in Lemma 2 to be able to bound the U-process with a Rademacher complexity which involves a sum of independent terms, we need to use such a construction for each fold I_l. Take the cross-fitting technique in Escanciano and Terschuur (2023) where we split $\{1,...,n\}$ into sets $\mathcal{C}=\{C_1,...,C_K\}$ and take the intersection between $\mathcal{C}^2$ and the set $\{(i,j)\in\{1,...,n\}^2 : i<j\}$. I_l can be either a triangle ($I_l\in T$, where $T=\{I_l : i\in C_f, j\in C_g, f<g, (i,j)\in I_l\}$) or a square ($I_l\in S$, where $S=\{I_l : i\in C_f, j\in C_g, f=g, (i,j)\in I_l\}$) and that in each case we can bound the U-process with the following Rademacher complexities

$$
\mathcal{R}_{n,l}(\Pi_{ab})=\begin{cases}\mathbb{E}_\varepsilon\left(\sup_{\pi\in\Pi}\left||C_k|^{-1}\sum_{i=1}^{|C_k|}\varepsilon_i\hat{R}^{(q)}_{\rho(i,k),|C_k|+i}\pi_{ab}(X_{\rho(i,k)},X_{|C_k|+i})\right|\right) & \text{if } I_l\in S\\ \mathbb{E}_\varepsilon\left(\sup_{\pi\in\Pi}\left|\lfloor|C_k|/2\rfloor^{-1}\sum_{i=1}^{\lfloor|C_k|/2\rfloor}\varepsilon_i\hat{R}^{(q)}_{i,\lfloor|C_k|/2\rfloor+i,l}\pi_{ab}(X_i,X_{\lfloor|C_k|/2\rfloor+i})\right|\right) & \text{if } I_l\in T,\end{cases}
$$

for $q=1,3$. Hence, by Lemmas 2 and 3 we have that

$$
\begin{aligned}
\binom{n}{2}^{-1}|I_l|\mathbb{E}&\left(\sup_{\pi\in\Pi_n}\left||I_l|^{-1}\sum_{(i,j)\in I_l}\hat{R}^{(1)}_{ij,l}\pi_{ab}(X_i,X_j)-\mathbb{E}[\hat{R}^{(1)}_{ij,ab,l}\pi_{ab}(X_i,X_j)|N_l^c]\right|\,\middle|N_l^c\right)\\
&=\mathcal{O}\left(\sqrt{\frac{S^{(1)}_{ab,l}VC(\Pi_{ab,n})}{\lfloor C_k/2\rfloor}}\right),
\end{aligned}
$$

where $S^{(1)}_{ab,l}=\mathbb{E}[\hat{R}^{(1)^2}_{ij,l}|N_l^c]$. Noting that $\mathbb{E}[S^{(1)}_{ab,l}]=\mathbb{E}[(m_{ab}(Z_i,Z_j,\hat{\gamma}_l,\varphi)-m_{ab}(Z_i,Z_j,\gamma,\varphi))^2]$

and using Assumption 4, Jensen's inequality, the fact that $|I_l| = |C_k| \times |C_m|$ if $I_l = I(C_k, C_m)$ and $|I_l| = |C_k| \times |C_k - 1|/2$ if $I_l = I(C_k, C_k)$ and that for evenly sized folds $|C_k|/(n-1) \leq 1$ for all $k = 1, ..., K$ we have that

$$\mathbb{E}\left(\sup_{\pi\in\Pi_n}\left|\binom{n}{2}^{-1}\sum_{(i,j)\in I_l}\hat{R}^{(1)}_{ij,l}\pi_{ab}(X_i,X_j) - \mathbb{E}[\hat{R}^{(1)}_{ij,ab,l}\pi_{ab}(X_i,X_j)|N_l^c])\right|\right)$$
$$= \mathcal{O}\left(\sqrt{VC(\Pi_{ab,n})\frac{a((1-K^{-1})n)^2}{n^{1+2\lambda_\gamma}}}\right).$$

The same bound applies by using the same arguments when we replace $\hat{R}^{(1)}_{ij,l}$ by $\hat{R}^{(3)}_{ij,l}$. This bound applies to all folds I_l, so, summing across all folds gives us the same asymptotic bound. Hence, we have bounded the first term on the right-hand side in (†). For the second term, we follow the same steps as with the first term to get the same bounds with λ_γ replaced by λ_φ. For the third term in (†), by Assumption 5 (i) (global robustness of α), $\mathbb{E}[\hat{R}^{(4)}_{ij,ab,l}|N_l^c] = \mathbb{E}[\hat{R}^{(6)}_{ij,ab,l}|N_l^c] = 0$. Thus, we can apply Lemmas 2 and 3 directly to get that for $q = 4, 6$

$$\mathbb{E}\left(\sup_{\pi\in\Pi_n}\left|\binom{n}{2}^{-1}\sum_{(i,j)\in I_l}\hat{R}^{(q)}_{ij,l}\pi_{ab}(X_i,X_j)\right|\right) = \mathcal{O}\left(\sqrt{VC(\Pi_{ab,n})\frac{a((1-K^{-1})n)^2}{n^{1+2\lambda_\alpha}}}\right).$$

Finally, the bound for the last term in (†) follows directly from Assumption 6

$$\mathbb{E}\left(\sup_{\pi\in\Pi_n}\left|\binom{n}{2}^{-1}\sum_{(i,j)\in I_l}(\hat{\xi}_{ij,l} + \hat{\xi}^{\gamma}_{ij,l} + \hat{\xi}^{\varphi}_{ij,l})\pi_{ab}(X_i,X_j)\right|\right) = \mathcal{O}\left(\frac{a(1-K^{-1})}{\sqrt{n}}\right).$$

Putting everything together we know that

$$
\begin{aligned}
\sqrt{n}\mathbb{E}\Big[\sup_{\pi\in\Pi_n}|\hat{W}_n(\pi)-\widetilde{W}_n(\pi)|\Big] &= \mathcal{O}\Bigg(\sqrt{VC(\Pi_{ab,n})\frac{a((1-K^{-1})n)^2}{n^{2\lambda_\gamma}}}\Bigg) \\
&+\mathcal{O}\Bigg(\sqrt{VC(\Pi_{ab,n})\frac{a((1-K^{-1})n)^2}{n^{2\lambda_\varphi}}}\Bigg) \\
&+\mathcal{O}\Bigg(\sqrt{VC(\Pi_{ab,n})\frac{a((1-K^{-1})n)^2}{n^{2\lambda_\alpha}}}\Bigg) \\
&+\mathcal{O}\Big(a(1-K^{-1})\Big)+o(1) \\
&=\mathcal{O}\Bigg(a((1-K^{-1})n)\Bigg(1+\sqrt{\frac{VC(\Pi_{ab,n})}{n^{2\min(\lambda_\gamma,\lambda_\varphi,\lambda_\alpha)}}}\Bigg)\Bigg).
\end{aligned}
$$

■

Proof of Theorem 1: Follows from Lemmas 3, 4 and 7. ■

Proof of Corollary 1: Let Γ_{ij}^{ab} and $\hat{\Gamma}_{ij,l}^{ab}$ be defined as in the Intergenerational mobility example and let

$$
K(\pi)=\mathbb{E}\Bigg[\sum_{(a,b)\in\{0,1\}^2}\Gamma_{ij}^{ab}\pi_{ab}(X_i,X_j)\Bigg],\quad \widetilde{K}_n(\pi)=\binom{n}{2}^{-1}\sum_{i<j}\Bigg[\sum_{(a,b)\in\{0,1\}^2}\Gamma_{ij}^{ab}\pi_{ab}(X_i,X_j)\Bigg],
$$

$$
\hat{K}_n(\pi)=\binom{n}{2}^{-1}\sum_{l=1}^{L}\sum_{(i,j)\in I_l}\Bigg[\sum_{(a,b)\in\pi}\hat{\Gamma}_{ij,l}^{ab}(Z_i,Z_j,\hat{\gamma}_l,\hat{\varphi}_l,\hat{\alpha}_l)\pi_{ab}(X_i,X_j).\Bigg].
$$

Note also that $W(\pi)=-|K(\pi)-t|$. Hence, we can write the regret as

$$
\mathbb{E}\Big[sup_{\pi\in\Pi_n}-|K(\pi)-t|+|K(\hat{\pi})-t|\Big]\le\mathbb{E}\Big[\sup_{\pi\in\Pi_n}|K(\pi)-K(\hat{\pi})|\Big].
$$

The result follows from applying Theorem 1 with W replaced by K. ■

C Multiple Integer Representation of Linear Rules

Here I adapt the multiple integer representation (MILP) of linear rules in Kitagawa and Tetenov (2018) to the U-statistic setting. The optimization problem is generally

$$
\hat{\pi}=\arg\max_{\pi\in\Pi}\sum_{i<j}\sum_{(a,b)\in\{0,1\}^2}g_{ij}^{ab}\pi_{ab}(X_i,X_j), \tag{8.3}
$$

for some weights g_{ij}^{ab}. For instance, $g_{ij}^{ab} = \hat{\Gamma}_{ij,l(ij)}^{ab}$, where $l(ij)$ is the fold that observation pair (i,j) belongs to. For $B \subset \mathbb{R}^{k+1}$ and $\beta \in B$, adding the constant to X_i, consider linear rules $\pi_\beta(X_i) = 1(X_i'\beta \geq 0)$. For a fixed (i,j), we have four terms

$$\begin{aligned} &g_{ij}^{11}1(\pi_\beta(X_i)=1)1(\pi_\beta(X_j)=1) + g_{ij}^{10}1(\pi_\beta(X_i)=1)1(\pi_\beta(X_j)=0) \\ &+g_{ij}^{01}1(\pi_\beta(X_i)=0)1(\pi_\beta(X_j)=1) + g_{ij}^{00}1(\pi_\beta(X_i)=0)1(\pi_\beta(X_j)=0). \end{aligned}$$

We can rewrite a MILP optimization problem equivalent to (8.3) as

$$\begin{aligned} \arg \max_{\beta\in B,(z_i,z_j,z_{ij})_{i<j}} &\sum_{i<j} g_{ij}^{11}z_{ij} + g_{ij}^{10}(z_i - z_{ij}) + g_{ij}^{01}(z_j - z_{ij}) + g_{ij}^{00}(1 - z_i - z_j + z_{ij}), \\ \text{s.t. } & z_{ij} \leq z_i, \quad z_{ij} \leq z_j, \quad z_{ij} \geq z_i + z_j - 1, z_{ij} \in [0,1], \quad \forall i<j, \\ & \frac{X_i'\beta}{C_i} < z_i \leq 1 + \frac{X_i'\beta}{C_i}, \quad z_i \in \{0,1\}, \quad \forall i = 1,...,n, \end{aligned}$$

where $C_i \geq \sup_{\beta\in B}|X_i'\beta|$. The constraints involving z_{ij} ensure that $z_{ij} = z_i z_j$ but avoid integer constraints on z_{ij}. The constraints involving z_i ensure that $z_i = 1(X_i'\beta \geq 0)$ as in Kitagawa and Tetenov (2018). Extensions to multiple linear rules or capacity constrains follow similarly as in Kitagawa and Tetenov (2018) once this quadratic structure is taken into account.

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